A Shock-Geometry Pre-Pass for Supersonic Rocket Aerodynamic Prediction and Flight Validation

Aidan Yu*

Independent Researcher

OpenRocket, the most widely used open-source rocketry flight simulator, is limited to subsonic aerodynamics. We extend its semi-empirical Barrowman framework to compressible flight through a once-per-timestep *shock-geometry pre-pass* that resolves local post-shock flow along the airframe and exposes it to each downstream drag, normal-force, and stability calculator. The pre-pass unifies classical compressible methods—Taylor–Maccoll cone flow, Prandtl–Meyer expansion, shock relations, Van Driest II skin friction, and DATCOM fin wave drag—in one open implementation, and twenty component models are externally benchmarked against published references. The simulator is validated against a 25-flight ground-truth corpus spanning Mach 0.54 to 4.33: the mean signed apogee error is -0.38% **(95% bootstrap confidence interval** $[-2.41, +1.72]$**, root-mean-square error** 5.34%**), and all 25 flights fall within** $\pm10\%$ **of measured apogee. On a decontaminated 12-flight blind holdout the model is *more* accurate (mean absolute error** 3.95%**) than on the 13 development flights (**5.47%**), so the two corpus-frozen base-drag constants generalize rather than overfit. Against RASAero II on the 25 paired flights there is no significant difference in absolute error (Wilcoxon** $p = 0.615$**). High-Mach prediction is exploratory: of 20 historical flights only three fall within** $\pm10\%$**. All code, data, and analysis are released for reproduction.**

Nomenclature

Symbol	Meaning
a	speed of sound, $\mathrm{m\,s^{-1}}$
A, B	Van Driest II compressibility auxiliaries, dimensionless
A_e	nozzle exit area, $\mathrm{m^2}$
A_{plan}, A_{ref}	body planform area; reference (cross-sectional) area, $\mathrm{m^2}$
AR	fin aspect ratio

*Independent Researcher; corresponding author, aidansyu@gmail.com; ORCID 0009-0005-9589-5314.

Symbol	Meaning
C_D	total axial-force (drag) coefficient
$C_{d,c}$	crossflow (Jorgensen) drag coefficient
C_f	compressible skin-friction coefficient
$C_{f,\text{inc}}$	incompressible (reference) skin-friction coefficient
C_N	normal-force coefficient
C_{N_α}	normal-force-curve slope, rad^{-1}
$C_{n,p\alpha}$, $C_{y,p\alpha}$	Magnus yawing-moment and side-force derivatives, rad^{-2}
C_p, $C_{p,\max}$	pressure coefficient; stagnation maximum (Rayleigh pitot)
C_{m_q}, $C_{m_{\dot\alpha}}$	pitch-damping derivatives, rad^{-1}
f	nose fineness ratio (length-to-diameter)
F	wall-to-edge temperature ratio, T_w/T_e, dimensionless
F_c, F_θ, F_x	Van Driest II skin-friction, momentum-thickness, and length transformation factors
K	DATCOM fin-section wave-drag shape factor
K_1, K_2, K_3	Barrowman/DATCOM fin normal-force coefficients
K_{cf}	body crossflow lift multiplier
K_{shape}	Dahlem–Buck nose-shape wave-drag factor
K_v	high-incidence body-vortex strength coefficient ($K_v = 0.20$)
K_{WB}, K_{BW}	fin-on-body and body-on-fin interference factors
L_{ref}	reference length (moment normalization), m
M, M_1, M_2	Mach number: freestream; pre-shock; post-shock (normal shock)
M_c	crossflow Mach number, $M\lvert\sin\alpha\rvert$
M_e	boundary-layer edge Mach number
M_s	local post-shock (fin-station) Mach number
m	compressibility parameter, $(\gamma - 1)M_e^2/2$
p	static pressure, Pa
p_{01}, p_{02}	pre-shock and post-shock stagnation pressure, Pa
r	turbulent recovery factor ($r = 0.88$)

Symbol	Meaning
R	specific gas constant for air (287.053 J/(kg K))
Re_L, Re_x	length-based and local Reynolds number
S	Sutherland constant ($S = 110.4$ K)
T, T_0, T_w	static, stagnation, and wall temperature, K
T_e, T_{aw}	boundary-layer edge and adiabatic-wall temperature, K
t/c	fin streamwise thickness-to-chord ratio
V_r, V_ϕ	radial and angular velocity components (Taylor–Maccoll), nondimensional
x_{CP}, x_{CG}	center of pressure; center of gravity, m
α	angle of attack, rad
β	Prandtl–Glauert/Ackeret compressibility factor $\sqrt{\lvert M^2 - 1\rvert}$ *and* oblique-shock wave angle (dual use), rad
γ, γ_{eff}	ratio of specific heats; effective (vibrationally relaxed) value
δ	local boundary-layer thickness, m
θ, θ_c	flow-deflection (wedge/ramp) angle; cone half-angle, rad
ϕ	Taylor–Maccoll polar (ray) angle, rad
Λ_{LE}	fin leading-edge sweep angle, rad
μ	dynamic viscosity, Pa s
$\nu(M)$	Prandtl–Meyer function, rad
τ	fin thickness-to-chord ratio (transonic similarity), $\tau = t/c$

I. Introduction

A. Background and Motivation

Sounding rockets, supersonic missile testbeds, and student-built research vehicles routinely traverse Mach 1 to Mach 5 during boost and coast. Mission designers in this class need trajectory tools that predict apogee, peak dynamic pressure, peak Mach number, and static and dynamic stability across the full ascent profile with engineering accuracy, and increasingly they need those tools to be open so that the underlying coefficient models can be audited, extended, and reproduced. The two most widely used open-source rocket trajectory simulators, OpenRocket [1] and RocketPy [2], both descend from the Barrowman engineering aerodynamic method [3] and are reliable below approximately Mach 1.1.

Above the sonic line the underlying constant-coefficient lookup tables, the incompressible skin-friction transformation, and the absence of post-shock local-flow corrections cause the predicted drag and stability derivatives to diverge from measured behavior, frequently producing apogee errors well in excess of twenty percent on high-supersonic and hypersonic sounding-rocket missions.

Commercial software fills part of this gap. RASAero II [4] couples a six-degree-of-freedom integrator to a Missile DATCOM-style coefficient generator and is widely used in the high-power and university-class rocketry communities. Its source code is closed and its underlying coefficient model is not externally auditable by the user. The semi-empirical lineage typified by Missile DATCOM and the Aeroprediction code family supports missile design at industrial scale [5, 6], and cross-comparisons such as that of Sooy and Schmidt against measured data [7] document its maturity; but that lineage is likewise closed and proprietary. No open-source rocket trajectory simulator presently offers externally benchmarked supersonic aerodynamic prediction coupled to an end-to-end, reproducible flight-corpus validation.

The objective of the present work is to close that gap within the OpenRocket fork OpenRocket-Plus: a single open-source, six-degree-of-freedom trajectory simulator whose aerodynamic subsystems are each traceable to a published wind-tunnel, ballistic-range, free-flight, or computational fluid-dynamics benchmark, and whose integrated apogee predictions are validated against an externally selected corpus of instrumented flights spanning the supersonic envelope. We validate end-to-end accuracy to Mach 4.33 against flight data and, separately and exploratorily, assess how far the same method reaches into the hypersonic regime on historical vehicles.

B. Prior Work

Barrowman's master's thesis [3] established the lift and center-of-pressure framework that anchors essentially every open-source rocket simulator written since 1967. Niskanen [1] implemented and extended that framework in OpenRocket, adding component-level drag bookkeeping and a Mach-dependent base-drag table valid through the transonic peak. RocketPy [2] provides a Python-language alternative also rooted in the Barrowman engineering pipeline, with comparable subsonic fidelity and a similar transonic ceiling. Both tools treat the freestream Mach number as the governing aerodynamic state uniformly along the body, an approximation that degrades as upstream shocks and expansions begin to process the local flow.

Rogers' RASAero II [4] couples a Missile DATCOM-style coefficient generator to a six-degree-of-freedom trajectory integrator and remains the most widely used non-OpenRocket prediction tool in high-power and university rocketry. The closed-source semi-empirical lineage against which the present work positions itself, Missile DATCOM and the Aeroprediction series [5–7], handles post-shock local conditions internally as part of its coefficient generation but exposes neither its source nor those intermediate states to the user. Figure 1 places these methods in a common hierarchy: the present work occupies the open-source engineering tier between the subsonic Barrowman/OpenRocket pipeline and the closed-source industrial Missile DATCOM/Aeroprediction codes, extending auditable open-source prediction across

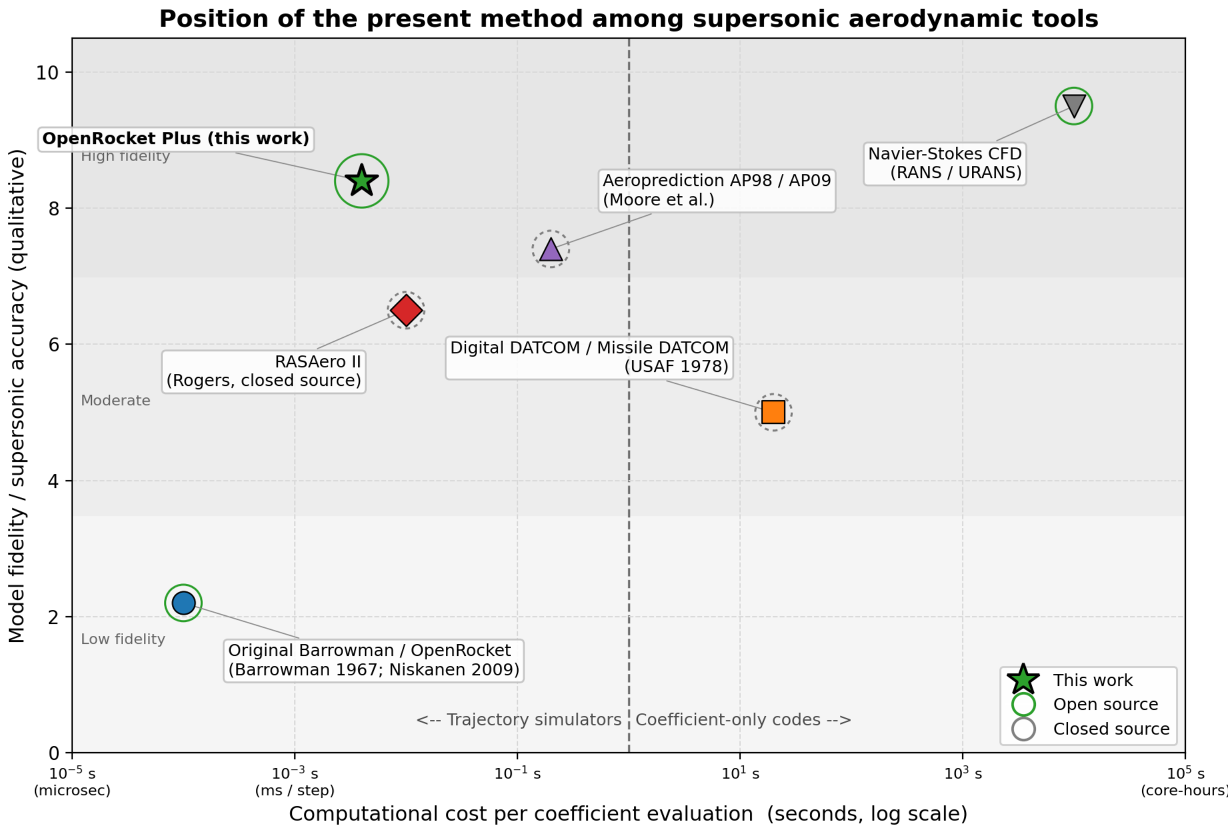

Fig. 1 Hierarchy of aerodynamic prediction methods for rocket trajectory simulation. The present work (OpenRocket-Plus) occupies the open-source engineering tier between the subsonic Barrowman/OpenRocket pipeline and the closed-source Missile DATCOM/Aeroprediction lineage, extending externally benchmarked open-source prediction across the supersonic envelope.

the supersonic envelope.

On the open-source-software side, recent precedent in this journal supports the publication of validated open aerospace software. The arcjetCV study [8] established that open-source aerospace software accompanied by a documented validation campaign and a permanent code archive is within the scope of the *Journal of Spacecraft and Rockets*, and a 2025 study of a low-cost amateur roll-control system [9] established that supersonic amateur-rocket hardware work anchored on open-source trajectory simulation fits within that scope as well. The present work combines those two precedents: open-source supersonic aerodynamics with a reproducible flight-corpus validation.

C. Gap Statement

The combined prior literature leaves three concrete gaps. First, no open-source rocket trajectory simulator distributes post-shock local-flow conditions to its component-level aerodynamic calculators; freestream Mach is applied uniformly along the body, which is increasingly incorrect above approximately Mach 1.5. Second, the engineering submodels underlying OpenRocket and RocketPy, including incompressible skin friction, $\cos^2 \Lambda_{LE}$ Ackeret fin wave drag, constant-coefficient base drag, and the absence of hypersonic blending, remain largely unchanged from their 1960s and 1970s antecedents and lack independent benchmarks above the transonic peak. Third, no open-source aerodynamic code has been subjected to end-to-end integrated flight-corpus validation against an externally selected set of instrumented supersonic flights, with the residual decomposed into bias and variance terms and compared on a paired basis against a commercial baseline.

D. Contributions

The present work addresses each gap with one corresponding contribution.

1) **A shock-geometry pre-pass architecture.** A once-per-timestep nose-to-tail surface march publishes post-shock local conditions, namely local Mach number, static pressure, and static temperature, to every component-level drag and stability calculator downstream of a single shared seam. The pre-pass is inert at subsonic Mach and is verified bit-for-bit against the Taylor–Maccoll cone solution and the Prandtl–Meyer function at body shoulders. We are explicit about what this contribution does and does not buy: its value is *local-flow fidelity* (correct post-shock conditions for fin loads and stability) and an architectural seam that enables the downstream supersonic submodels, rather than a large change in integrated apogee, which is dominated by lower-Mach drag and to which the pre-pass contributes only a fraction of a percentage point.
2) **An externally benchmarked supersonic subsystem replacement.** Each underlying Barrowman-era engineering submodel is replaced and independently benchmarked against published wind-tunnel, ballistic-range, free-flight, or computational fluid-dynamics data, with the governing equation, regime of applicability, and mean absolute percent error reported per subsystem. Where a model decision is corpus-calibrated rather than literature-anchored, we disclose it as such.
3) **A reproducible flight-corpus validation.** An externally selected corpus of 25 instrumented flights spanning Mach 0.54 to 4.33 enables end-to-end integrated validation of the present method and a direct paired comparison against RASAero II. On this corpus OpenRocket-Plus achieves a mean signed apogee error of -0.38% with all 25 flights within $\pm 10\%$ of measured altitude; the corresponding RASAero II predictions, as recorded by Rogers, give a mean signed error of $+2.46\%$ with 22 of 25 within $\pm 10\%$. The paired difference in absolute error is not statistically significant (Wilcoxon signed-rank $p = 0.615$; 95% bootstrap confidence interval on the mean paired difference $[-2.16, +0.96]$ percentage points, straddling zero), so the honest claim is *parity* with this version-locked commercial baseline, not superiority. The corpus, analysis scripts, and source code are released openly under a permanent digital object identifier [10].

Two scope qualifications frame these contributions honestly. The headline validation corpus was selected externally by Rogers for an independent RASAero II comparison rather than curated by the present authors, so its accuracy statistics are an outcome-independent validation result; the recorded RASAero II values are Rogers' own predictions and are not fresh, independently rerun pre-flight estimates. Separately, although the underlying compressible-flow infrastructure demonstrates a method capability reaching Mach 7, we do *not* claim hypersonic flight validation in the headline: high-Mach behavior is assessed exploratorily on a full, uncurated set of historical sounding rockets, where motor and geometry reconstruction uncertainty dominates the residual.

E. Paper Organization

Section II presents the shock-geometry pre-pass architecture and establishes the analytical-reference verification that anchors the novelty claim. Section III documents the underlying compressible-flow infrastructure: atmosphere, viscosity, the effective ratio of specific heats, the smoothed compressibility factor through Mach 1, and the closed-form shock and expansion relations. Sections IV and V catalog the drag and stability submodels respectively, each with its governing equation, regime of applicability, and published benchmark. Section VI rolls up the externally benchmarked subsystems with explicit verification-and-validation methodology and discloses the corpus-calibrated model decisions. Section VII validates against four primary published computational fluid-dynamics comparators, with a fifth study as an independent corroborating source. Section VIII reports the headline result: end-to-end integrated validation against the 25-flight corpus, with bias–variance decomposition, per-regime breakdown, paired RASAero II comparison, and distribution and normality characterization, together with the decontaminated prospective holdout that addresses the partly in-sample nature of two calibrated constants. Section IX reports the exploratory high-Mach sounding-rocket assessment over the full historical set, presented without cherry-picking. Section X discloses known limitations with quantitative magnitude and root cause where identified, including a four-flight operational sensitivity sweep, and Section XI concludes with prioritized future work.

II. Shock-Geometry Pre-Pass Architecture

This section presents the architectural contribution that distinguishes the present work from prior Barrowman-pipeline open-source simulators. The shock-geometry pre-pass is a once-per-timestep nose-to-tail surface march that publishes the post-shock local Mach number, static pressure, and static temperature at every component station before any downstream drag or stability calculator runs. Subsequent sections consume those local conditions through a stable query interface. The architectural value of the pre-pass lies in local post-shock fidelity and in providing a single seam through which all of the supersonic submodels of Secs. III–V are driven; as quantified in Section II.G, its contribution to *integrated* apogee is small, because apogee integrates a trajectory whose drag impulse is dominated by lower-Mach flight.

A. Motivation

The freestream Mach number governs aerodynamic-coefficient behavior only when the flow over each component is unprocessed by upstream shocks and expansions. On a slender finned rocket at moderate supersonic Mach this assumption breaks at the first body shoulder: the conical or ogival nose shock processes the flow before it reaches the cylindrical afterbody, the fins, and the base. A 15° cone at freestream Mach 2.5, for instance, leaves the post-shock surface Mach near 2.16—a fourteen-percent reduction. Because the Prandtl–Glauert compressibility factor $\beta = \sqrt{|M^2 - 1|}$ is nonlinear, that fourteen-percent Mach reduction translates to roughly an eighteen-percent change in the fin lift-curve-slope coefficient $K_1 = 2/\beta$. Similar effects propagate through fin wave drag, fin–body interference, and base-region inflow conditions. Evaluating every downstream coefficient at the freestream Mach therefore introduces a systematic bias in the *local* aerodynamic loads, even where its effect on the integrated trajectory is modest.

Closed-source missile aerodynamic codes such as Missile DATCOM and the Aeroprediction lineage [5–7] handle post-shock local conditions internally as part of their coefficient generators but do not expose them as a shared architectural seam. Open-source rocket simulators do not address the local-flow question at all: OpenRocket and RocketPy evaluate every component at the freestream Mach [1, 2]. The present pre-pass makes the post-shock state an explicit, queryable object shared by every downstream calculator within an open six-degree-of-freedom trajectory simulator.

B. Data flow

The pre-pass executes once per call to the top-level aerodynamic-forces routine, immediately after the `FlightConditions` object is populated and before any per-component calculator runs. It produces a `ShockGeometry` object indexed by axial station; each downstream calculator queries that object for its local post-shock state. The data path is summarized below:

```
function getAerodynamicForces(rocket, fc):
    shock_geom = ShockGeometry.compute(rocket, fc)
    forces = {}
    for component in rocket.components():
        local = shock_geom.queryAt(component.axialStation())
        forces[component] = calc(component, fc, local)
    return aggregate(forces)
```

The pre-pass is configuration-aware: it invalidates and recomputes on geometric changes (staging, fairing separation, side-booster jettison) but is otherwise pure with respect to `FlightConditions`. Below the sonic line the surface march short-circuits to a pass-through that returns freestream conditions at every station; the measured subsonic overhead is below ten microseconds per call on the reference hardware (single-threaded Intel Core i7), which is below the

Table 2 `ShockGeometry` calculator input/output contract. Subsonic column gives the pass-through value returned below $M_\infty = 1.0$.

Field	Direction	Description	Subsonic value
`Rocket` configuration	in	Component tree, axial stations, half-angles, shoulder/boat-tail flags	passed through
`FlightConditions.machNumber`	in	Freestream Mach number M_∞	passed through
`FlightConditions.aoa`	in	Vehicle angle of attack (windward-side bias when present)	unused
`localMach[station]`	out	Post-shock local Mach at each axial station	returns M_∞
`localPressure[station]`	out	Post-shock static-pressure ratio p/p_∞	returns 1.0
`localTemperature[station]`	out	Post-shock static-temperature ratio T/T_∞	returns 1.0
`shockStandoff`	out	Detached-shock standoff distance for blunt noses	∞ (no shock)
`activationBlend`	out	C^1 blend weight $w \in [0, 1]$ through $M = 1.0$–1.1	returns 0.0

measurement floor of the simulator's per-timestep budget. The full input/output contract is given in Table 2.

Figure 2 shows the data-flow block diagram.

C. Surface marching

The march initializes at the nose tip. For a sharp conical nose the inflow is processed through a Taylor–Maccoll cone solution [11] evaluated at the local tip half-angle, returning the cone-surface Mach and pressure as the seed state. For an ogival nose the tip slope is read as an instantaneous cone half-angle, and the Taylor–Maccoll seed is followed by a Prandtl–Meyer expansion at each subsequent strip increment. For a blunt nose a detached normal-shock solution seeds the post-shock state, and the surface march resumes around the shoulder.

Body-station traversal walks downstream along the axisymmetric body, applying at each surface element either a Prandtl–Meyer expansion (for a contracting turning angle, where the surface inclination decreases relative to the freestream) or an oblique-shock turn (for an expanding turning angle, where the surface inclination increases). The branch is selected by the sign of the turning angle relative to the prior surface tangent; turning angles below a numerical threshold (10^{-9} rad) are skipped to preserve the isentropic state. Boat-tails and aft-body contractions are handled by the expansion branch; shoulders, fin-leading-edge regions, and forward conical transitions invoke the oblique-shock branch.

Pressure and temperature ratios accumulate multiplicatively along the march; Mach is updated through the Prandtl–Meyer function [12] for expansions and through the post-shock θ–β–M relation for compressions. The iterative solvers—a Taylor–Maccoll Runge–Kutta-4 integration with 500 angular steps, an oblique-shock bisection on the shock angle β, and a Prandtl–Meyer Newton–Raphson inversion—converge to a 10^{-12} relative tolerance.

D. Local-condition query interface

Downstream calculators receive the pre-pass output through a stable query interface keyed on axial station. The body calculator reads `localMach` and `localPressure` at each body-element centroid for the shock-expansion strip integral and the crossflow body lift. The fin-set calculator reads `localMach` at the fin axial station and uses it in the

DATCOM $K_1/K_2/K_3$ fin normal-force computation and in the DATCOM 4.1.5.1 fin wave-drag computation. The fin–body interference factors and the shock-wave/boundary-layer-interaction chord-reduction model draw from the same interface, and the base-drag calculator reads the post-shock local pressure for the base dynamic-pressure rescaling. A single seam thus supplies the local flow state to every supersonic submodel of Secs. III–V.

E. Numerical guards and C^1 activation

The pre-pass is gated by a continuous activation weight $w(M_\infty)$ that ramps from zero at $M_\infty = 1.0$ to one at $M_\infty = 1.1$ via a smoothstep,

$$w(M_\infty) = 3t^2 - 2t^3, \qquad t = \frac{M_\infty - 1.0}{0.1}, \quad t \in [0, 1]. \tag{1}$$

Below $M_\infty = 1.0$ the pre-pass returns freestream conditions everywhere; above $M_\infty = 1.1$ it returns the full post-shock state; through the activation band each component's local state is the smoothstep blend of the two. The activation band is C^1-continuous in M_∞, which preserves the smoothness of the assembled drag and stability coefficients required by the fixed-step Runge–Kutta-4 trajectory integrator.

Additional guards include a minimum surface-element length below which the expansion/oblique-shock branch is suppressed (avoiding accumulated round-off on geometries with very fine longitudinal discretization); a hard fallback to freestream conditions when the computed local Mach descends below 1.0 due to repeated expansion (preserving physical consistency); and bounds-checking on the post-shock θ–β–M solution to ensure $M_2 > 0$ for all admissible inputs.

F. Verification against analytical references

The pre-pass was verified against analytical references at six combinations of freestream Mach and cone half-angle spanning $M_\infty = 1.5$, 2.5, and 5.0 at $\theta_c = 10°$ and 20°. Cone-surface Mach reproduced the Taylor–Maccoll exact solution to 0.00% in all six cases, the residual falling below the eight-digit print precision of the diagnostic; the underlying Taylor–Maccoll shock-angle solver matches the tabulated cone solution to within 0.825% (against a 1% acceptance gate). Shoulder-expansion turning reproduced the Prandtl–Meyer relation [12] to better than 4×10^{-11}% at all tested turning angles between 1° and 25°, and the oblique-shock branch reproduced the closed-form turn relation to a maximum error of 0.021%. Figure 3 plots the surface-Mach march for the six reference cases against the analytical envelope. This verification level is consistent with the AIAA Editorial Policy on Numerical and Experimental Accuracy [13], which directs engineering codes to anchor against analytical or tabulated solutions where available; the pre-pass is reported as an externally benchmarked subsystem in the Sec. VI roll-up.

G. Contribution to integrated apogee: ablation

The architectural prominence of the pre-pass should not be confused with a large effect on integrated apogee. To quantify its contribution honestly, each supersonic mechanism was disabled in isolation and the resulting change in predicted apogee was measured across 24 of the 25 corpus flights (the MESOS two-stage closure requires custom-motor preload and is exercised separately). Disabling the shock-geometry pre-pass changes the integrated apogee error by a mean magnitude of only 0.15 percentage points, with a maximum of 3.6 percentage points on a single Mach-2.46 flight (FMJ BR6). By comparison, the externally calibrated finned-base drag augmentation is the dominant apogee driver, with a mean magnitude of 8.10 percentage points and a maximum of 39.5 percentage points; Van Driest II skin friction contributes 0.87 percentage points (mean) and matters chiefly at high Mach, and the DATCOM fin wave-drag term contributes 0.39 percentage points.

The small integrated-apogee effect has a straightforward cause: apogee is a drag-impulse integral over a trajectory whose drag is dominated by lower-Mach flight, and the pre-pass is inert below $M_\infty = 1.0$ by construction (Section II.E). The pre-pass is therefore *not* presented as a gross-apogee improvement. Its value is twofold: it supplies correct *local* post-shock conditions for fin loads and static stability—verified to the analytical references of Section II.F, where freestream evaluation would bias the local coefficients as shown in the motivation above—and it is the architectural seam through which the supersonic drag and stability models of Secs. III–V are driven. The integrated-apogee accuracy reported in Sec. VIII rests primarily on those downstream models and on the externally calibrated finned-base term, not on the pre-pass in isolation. The finned-base augmentation is identified here only as the dominant term motivating the present pre-pass framing; the dominance of the finned-base augmentation in the apogee error budget is examined in detail—as a distinct, deferred research question rather than a re-slice of the present integrated result—in a companion base-drag intercomparison study [14]. That dedicated intercomparison of the base-drag closures that govern integrated apogee—Chapman–Korst, ESDU 77021, the $0.064 + 0.186/M^2$ correlation, and the finned-base augmentation—is deferred to that companion paper, and the present work does not draw any conclusion ranking those closures.

III. Atmosphere, Compressibility, and Shock Relations

The models of the following sections inherit a common thermodynamic and compressible-flow infrastructure: the atmospheric profile, the temperature dependence of viscosity and the ratio of specific heats, the compressibility factor through the sonic line, and the closed-form shock and expansion relations. Every quantity in this section was verified against an analytical or authoritative tabulated reference before any downstream model was activated.

A. Speed of Sound and U.S. Standard Atmosphere

The thermodynamic speed of sound for dry air is

$$a = \sqrt{\gamma R T}, \qquad R = 287.053\,\mathrm{J/(kg\,K)}, \tag{2}$$

evaluated at the local static temperature. The original linear fit $a = 331.3 + 0.606\,(T - 273.15)$ errs by approximately 0.6 % at tropopause temperatures. The revised implementation evaluates Eq. (2) directly with $\gamma = 1.4$ in the freestream and an effective γ (Sec. III.C) above the vibrational threshold. Verification against the U.S. Standard Atmosphere 1976 [15] over twenty altitudes from sea level to 80 km gave a maximum error of 0.016 % (Fig. 4).

B. Sutherland Viscosity

Dynamic viscosity follows Sutherland's law

$$\mu(T) = \mu_{\mathrm{ref}} \left(\frac{T}{T_{\mathrm{ref}}}\right)^{3/2} \frac{T_{\mathrm{ref}} + S}{T + S}, \tag{3}$$

with $\mu_{\mathrm{ref}} = 1.716 \times 10^{-5}\,\mathrm{Pa\,s}$, $T_{\mathrm{ref}} = 273.15\,\mathrm{K}$, and $S = 110.4\,\mathrm{K}$. The legacy linear fit produced viscosity errors exceeding 50 % at post-shock wall temperatures, contaminating the skin-friction model. Equation (3) was verified against the standard reference air-property tables [16] over 150 K to 500 K with a mean absolute percent error (MAPE) of 0.54 % and no systematic bias (Fig. 5). The viscosity ratio implied by Eq. (3) also supplies the F_θ factor of the Van Driest II compressibility transformation (Sec. III.F).

C. Effective Ratio of Specific Heats

Above stagnation temperatures of approximately 800 K, vibrational excitation of N_2 and O_2 depresses the effective ratio of specific heats. A piecewise model interpolates between thermally perfect, vibrationally relaxed, and partially dissociated regimes:

$$\gamma_{\mathrm{eff}}(T_0) = \begin{cases} 1.400, & T_0 \le 800\ \mathrm{K}, \\ 1.400 - 7.5 \times 10^{-5}\,(T_0 - 800), & 800 < T_0 \le 2000\ \mathrm{K}, \\ 1.310 - 2.5 \times 10^{-5}\,(T_0 - 2000), & 2000 < T_0 \le 4000\ \mathrm{K}, \\ 1.250, & T_0 > 4000\ \mathrm{K}. \end{cases} \tag{4}$$

The break points correspond to vibrational-equilibrium onset (800 K), substantial vibrational excitation (2000 K), and incipient dissociation (4000 K) [17]. The model is piecewise, with a value step below 1 % at the 4000 K break point, far below the downstream sensitivity floor; flight time above $T_0 = 4000\,\mathrm{K}$ is negligible. Real-gas dissociation chemistry is out of scope for the vehicles considered here.

D. Smooth Compressibility Factor Through Mach One

The Prandtl–Glauert/Ackeret compressibility factor

$$\beta(M) = \sqrt{\left|1 - M^2\right|} \tag{5}$$

appears in nearly every supersonic coefficient closed form. The legacy hard clamp $\beta_{\min} = 0.25$ produced a flat plateau from $M \approx 0.97$ to 1.03 and was physically incorrect on both sides of the sonic line. The revised implementation embeds a cubic Hermite spline through the band $M \in [0.95, 1.05]$:

$$\beta(M) = \begin{cases} \sqrt{1 - M^2}, & M < 0.95, \\ H_3\big(M;\ M_L, M_H, \beta_L, \beta_H, \beta'_L, \beta'_H\big), & 0.95 \le M \le 1.05, \\ \sqrt{M^2 - 1}, & M > 1.05, \end{cases} \tag{6}$$

where H_3 is the cubic Hermite polynomial with endpoint values and slopes taken from the analytical expressions at $M_L = 0.95$ and $M_H = 1.05$. The spline is C^1 continuous, strictly positive (minimum 0.28 near $M = 1$), and asymptotes correctly to $\sqrt{M^2 - 1}$ above $M = 1.05$. The blending region is one of thirteen documented in Table 3; every regime transition in the present work obeys the same C^1 discipline because fourth-order Runge–Kutta integration is intolerant of jumps in C_D, C_{N_α}, or C_{m_q}.

E. Normal, Oblique, and Prandtl–Meyer Relations

Normal-shock jump conditions in a calorically perfect gas are

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}\left(M_1^2 - 1\right), \qquad M_2^2 = \frac{M_1^2 + 2/(\gamma - 1)}{2\gamma M_1^2/(\gamma - 1) - 1}. \tag{7}$$

The stagnation-pressure ratio implied by Eq. (7) yields the Rayleigh pitot formula used downstream for Modified Newtonian theory:

$$\frac{p_{02}}{p_{01}} = \left[\frac{(\gamma + 1)M_1^2}{(\gamma - 1)M_1^2 + 2}\right]^{\gamma/(\gamma-1)} \left[\frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1}\right]^{-1/(\gamma-1)}. \tag{8}$$

The oblique-shock solver solves the θ–β–M relation

$$\tan\theta = 2\cot\beta\, \frac{M_1^2 \sin^2\beta - 1}{M_1^2(\gamma + \cos 2\beta) + 2} \tag{9}$$

by bisection on the shock angle β between the Mach angle $\sin^{-1}(1/M_1)$ and 90°. For conical noses the Taylor–Maccoll equations [11]

$$\frac{dV_r}{d\phi} = V_\phi, \qquad \frac{dV_\phi}{d\phi} = \frac{V_\phi^2 V_r - \frac{\gamma-1}{2}\left(1 - V_r^2 - V_\phi^2\right)\left(2V_r + V_\phi \cot\phi\right)}{\frac{\gamma-1}{2}\left(1 - V_r^2 - V_\phi^2\right) - V_\phi^2} \tag{10}$$

are integrated by fourth-order Runge–Kutta with 500 steps, iterating on the shock angle until the radial velocity at the cone surface vanishes. The Prandtl–Meyer function

$$\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}}\,\arctan\sqrt{\frac{\gamma-1}{\gamma+1}\left(M^2-1\right)} - \arctan\sqrt{M^2-1} \tag{11}$$

is solved for the downstream Mach after a turning angle $\Delta\theta$ by Newton–Raphson with an analytic derivative. All iterative loops converge to a 10^{-12} relative tolerance.

F. Van Driest II Compressibility Transformation

The compressible turbulent skin-friction model used by the drag calculator of the following section maps an incompressible flat-plate friction coefficient to the compressible value through the Van Driest II transformation of Hopkins and Inouye [18]. Defining

$$m = \frac{\gamma-1}{2}\,M_e^2, \qquad F = \frac{T_w}{T_e}, \qquad r = 0.88, \tag{12}$$

where M_e is the boundary-layer edge Mach number, T_w/T_e is the wall-to-edge static-temperature ratio, and r is the turbulent recovery factor, the transformation introduces the auxiliary quantities

$$A = \sqrt{\frac{r\,m}{F}}, \qquad B = \frac{1 + r\,m - F}{F}, \tag{13}$$

and the compressibility factor

$$F_c = \frac{r\,m}{\left(\arcsin A_1 + \arcsin A_2\right)^2}, \qquad A_1 = \frac{2A^2 - B}{\sqrt{4A^2 + B^2}}, \qquad A_2 = \frac{B}{\sqrt{4A^2 + B^2}}. \tag{14}$$

The numerator $r\,m = T_{\mathrm{aw}}/T_e - 1$ is the recovery-temperature ratio deficit; for an adiabatic wall, $T_w/T_e = 1 + r\,m$, so that $B = 0$. The temperature-ratio factor F_θ is formed from the Sutherland viscosity ratio of Eq. (3), the Reynolds-number factor is $F_x = F_\theta/F_c$, and the compressible friction coefficient is $C_f = C_{f,\mathrm{inc}}/F_c$.

Table 3 Mach blending regions used in the present work; all transitions are C^1 continuous to maintain Runge–Kutta trajectory stability.

Physical quantity	Mach band	Method	Reference
Compressibility factor β	0.95–1.05	Cubic Hermite, value + slope matched	Sec. III.D
Skin friction C_f (incompressible $\rightarrow$ Van Driest II)	0.9–1.1	Polynomial interpolation	Sec. III.F
Base drag (transonic peak)	0.85–1.3	Degree-4 polynomial anchored to peak	Drag models
Chapman–Korst turbulent base drag	1.2–1.4	Smoothstep blend from supersonic correlation	Drag models
Chapman laminar base drag	1.3–2.5	Smoothstep blend from supersonic correlation	Drag models
Fin wave drag onset (zero $\rightarrow$ DATCOM 4.1.5.1)	0.9–1.2	Cubic Hermite	Drag models
Fin C_{N_α} ($K_1/K_2/K_3$)	0.9–1.5	Polynomial blend	Stability models
Transonic similarity (ESDU)	0.9–1.5	Universal $h(K_{\text{trans}})$ blend	Stability models
PNK fin–body interference	0.85–1.15	Smoothstep from $F = 1$	Stability models
Nose wave drag (tables $\rightarrow$ analytical)	1.3–1.5	Smoothstep	Drag models
Body C_{N_α}/CP supersonic shift	0.8–1.3	Smoothstep from Barrowman	Stability models
Modified Newtonian hypersonic blend	4.0–6.0	Smoothstep	Drag models
ShockGeometry activation	1.0–1.1	Smoothstep toward freestream	Sec. III.D

G. Verification Against NACA Report 1135

Each shock building block was verified against the tabulated values in NACA Report 1135 [12]. The normal-shock pressure ratio (Fig. 6), oblique-shock angle (Fig. 7), Prandtl–Meyer function (Fig. 8), and Rayleigh pitot $C_{p,\max}$ (Fig. 9) reproduced the reference tables to better than 0.1 % across $M = 1.5$–10 and 5° to 40° cone half-angles. Component checks gave a maximum error of 0.016 % for the speed of sound, 0.021 % for the oblique-shock angle, and 0.004° for the Prandtl–Meyer function. The Taylor–Maccoll cone-shock-angle solver achieved a maximum error of 0.825 % (gate 1 %) against the tabulated supersonic cone solutions of NASA SP-3004 [17, 19]; the surface pressure coefficient itself matched the exact analytical result to less than 0.01 %.

IV. Drag Models

The total axial-force coefficient is assembled from independently validated contributions, each evaluated at the locally corrected post-shock state supplied by the shock-geometry pre-pass of Sec. II:

$$C_D = C_{D,\,\text{friction}} + C_{D,\,\text{pressure}} + C_{D,\,\text{base}} + C_{D,\,\text{override}} + C_N \sin\alpha. \tag{15}$$

Each submodel below states its regime of applicability, governing equation, and the published benchmark against which it was assessed. Table 4 summarizes the inventory.

A. Nose and Body Wave Drag

For conical noses the wave-drag coefficient equals the Taylor–Maccoll surface pressure coefficient—the exact inviscid result for steady conical flow at zero incidence (Sec. III)—and serves as the reference for all shape-correction methods.

For tangent and secant ogives, parabolic, and shock-attached power-law noses, a shock-expansion strip integrator marches $N_{\text{strip}} = 100$ conical frustum strips from tip to base. Each strip applies a Prandtl–Meyer expansion or an oblique shock according to the local turning angle; pressure and temperature ratios accumulate multiplicatively. The pressure-drag integral is

$$C_{d,\,\text{wave}} = \frac{2}{R_{\text{aft}}^2 - R_{\text{fore}}^2} \sum_{i=1}^{N_{\text{strip}}} C_{p,i}\, r_{\text{mid},i}\, \Delta r_i, \tag{16}$$

summing only windward strips, with tip initial conditions seeded from Taylor–Maccoll at the local tip half-angle.

For Haack-series, parabolic, and selected power-law noses, the Dahlem–Buck correction [20] scales the equivalent-cone result by an empirical shape factor and a fineness correction,

$$C_{d,\,\text{wave}} = C_{d,\,\text{cone}}(M, \theta_{\text{equiv}})\, K_{\text{shape}} \left(\frac{3}{f}\right)^{1.6}, \tag{17}$$

with $K_{\text{shape}} = 1.00$ for cones, 0.85 for Lighthill–von Kármán (L-V) ogives, 0.88 for parabolics, 0.90–0.95 for power-law families, and 0.60 for the Haack series. A smoothstep over $M = 1.3$–1.5 fades the legacy NASA TR R-100 [21] tabulated drag into the analytical result above $M = 1.5$. Below the drag-divergence Mach $M_{\text{dd}} = 0.95 - 0.15 \sin^{0.4} \theta_{\text{tip}}$ wave drag is zero; above M_{dd} a C^1 cubic Hermite connects zero drag to the first analytical point. Above $M = 5$ the nose pressure crosses over to Modified Newtonian theory (Sec. IV.F). The combined model was validated against NACA RM A52H28 [22] wind-tunnel pressure measurements for five fineness-3 nose shapes (cone, quarter-power, three-quarter-power, Haack, and L-V ogive) at $M = 1.5$–3.0, giving an aggregate mean absolute error of 0.029 in C_d against an acceptance gate of 0.035 (Fig. 10).

B. Fin Wave Drag (DATCOM 4.1.5.1)

The fin wave-drag model replaces the legacy $\cos^2 \Lambda_{LE}$ Ackeret approximation with the full DATCOM 4.1.5.1 method [23], branching on whether the fin leading edge is subsonic or supersonic relative to the local post-shock Mach number:

$$C_{d,\,\text{wave}} = \begin{cases} \dfrac{K}{\beta} \left(\dfrac{t}{c}\right)^2, & \beta \cot \Lambda_{LE} \geq 1 \quad \text{(supersonic LE)}, \\[2ex] K \cot \Lambda_{LE} \left(\dfrac{t}{c}\right)^2, & \beta \cot \Lambda_{LE} < 1 \quad \text{(subsonic LE)}, \end{cases} \tag{18}$$

where $\beta = \sqrt{M^2 - 1}$, the section shape factor $K = 4.0$ for double-wedge (hexagonal) sections and $16/3$ for biconvex or rounded sections per DATCOM Table 4.1.5.1-A, and t/c is the streamwise thickness-to-chord ratio. A cubic Hermite blend over $M = 0.9$–1.2 fades zero wave drag into the full DATCOM result. The implementation reproduces the closed-form Ackeret formula on 15 analytical cases to 0.00% error (code verification) and was validated against free-flight measurements of a 60-deg delta wing from NACA TN 3650 [24], yielding a mean absolute percentage error of 21.0% across 12 points over $M = 1.1$–2.5 (Fig. 11). The residual reflects the difficulty of predicting wave drag near the subsonic-to-supersonic leading-edge transition.

C. Base Drag

Four submodels cover the base-drag regimes of a slender finned rocket. For $M > 1.3$ with a turbulent boundary layer the supersonic base-drag correlation

$$C_{d,\,\mathrm{base}} = 0.064 + \frac{0.186}{M^2} \tag{19}$$

is applied. This empirical form is the supersonic asymptote for slender axisymmetric afterbodies documented in ESDU 77021 [25], and it was validated here against NACA TN 3393 [26] at four turbulent points over $M = 2.73$–4.48, giving a mean absolute percentage error of 15.9% (Fig. 12). For perfect-finish vehicles with delayed transition the laminar base-pressure correlation of Chapman [27]

$$C_{p,\,b,\,\mathrm{lam}} = \frac{C_{\mathrm{lam}}}{M^2\sqrt{\mathrm{Re}_L}}, \qquad C_{\mathrm{lam}} = 1870, \tag{20}$$

is used, validated against the four laminar TN 3393 points with a mean absolute percentage error of 4.4%. Applying the turbulent form, Eq. (19), to the laminar data instead yields a mean absolute percentage error of 44%: the boundary-layer state at the base materially governs the closure. A Chapman–Korst free-shear-layer model with the ESDU 77021 [25] boundary-layer-thickness correction blends with Eq. (19) over $M = 1.2$–1.4. A Viswanath boattail correction [28] reduces base drag by 15–40% for typical boattail half-angles of 6–16 deg, and a power-on multiplier derived from the solid-rocket motor exit-flow guidance of NASA SP-8039 [29] reduces base drag during burn using the nozzle exit-area and pressure ratios. Below $M = 0.85$ a Hoerner subsonic correlation [30] is recovered, and the transonic peak near $M \approx 1.05$ is represented by a degree-4 polynomial anchored at the regime boundaries.

D. Skin Friction: Van Driest II

The compressible skin-friction coefficient uses the Van Driest II transformation of Eqs. (12)–(14) (Sec. III), in the Hopkins–Inouye formulation [18], replacing the legacy Eckert reference-temperature method. The transformed Reynolds number $F_x\,\mathrm{Re}_x$, with $F_x = F_\theta / F_c$, is substituted into the Karman–Schoenherr implicit relation

$$\frac{0.242}{\sqrt{C_{f,\,\mathrm{inc}}}} = \log_{10}\left(\mathrm{Re}_x\, C_{f,\,\mathrm{inc}}\right), \tag{21}$$

solved by fixed-point iteration seeded with Schlichting's $\log^{-2.58}$ approximation, and the compressible coefficient follows as $C_f = C_{f,\text{inc}}/F_c$. Hopkins and Inouye [18] demonstrated that Van Driest II gave the best agreement with experiment among candidate transformations over $M = 1.5$–9. At $M = 5$ the compressible C_f is approximately half of its incompressible value at matched length Reynolds number (Fig. 13); the Eckert reference-temperature method reproduces this trend to within roughly 10% over $M = 1.5$–9, and Van Driest II was retained for its slightly closer agreement with experiment at the upper end of this range [18].

E. Boundary-Layer Transition

The transition Reynolds number is Mach-dependent, with a laminar-fraction cap that prevents unphysically long laminar runs at high Mach. The low-subsonic crossover $\mathrm{Re}_{x,\mathrm{tr}} = 5 \times 10^5$ rises to approximately 5×10^6 at $M = 4$, consistent with the compressible-stabilization trend of the boundary layer. The laminar fraction is capped at 0.7 of the body length to preserve consistency with the post-shock entropy layer.

F. Hypersonic Blending

Above $M = 5$ the inviscid nose and body pressure distributions transition to Modified Newtonian theory, $C_p = C_{p,\max} \sin^2 \theta$, with $C_{p,\max}$ taken from the normal-shock stagnation pressure of Sec. III. A smoothstep weight $w = 3t^2 - 2t^3$ with $t = (M - 4)/2$ blends the shock-expansion result into the Newtonian asymptote over $M = 4$–6. The combined hypersonic model was assessed against DTIC AD0487365 [31] ballistic-range cone foredrag at $M = 6.5$–17.2 for 8-, 12-, and 16-deg half-angle cones, giving an aggregate mean absolute percentage error of 19.7% across 11 points (Fig. 14). The largest residual, +57%, occurs for the thinnest 8-deg cones at the lowest-Reynolds-number row, where friction and base drag dominate the foredrag and the reference boundary-layer state is incompletely specified; the blunter 16-deg cone agrees most closely. These results are reported as exploratory: the headline flight validation (Sec. VIII) extends only to $M = 4.33$, and the hypersonic closure is a demonstrated method capability rather than a flight-validated claim.

V. Stability and Dynamic Stability Models

The static and dynamic stability calculators consume the same post-shock local-flow conditions produced by the shock-geometry pre-pass as the drag model of the preceding section. Each submodel is summarized below with its regime of applicability, its departure from the legacy Barrowman formulation, and the published benchmark against which it is validated. The static-stability chain (Secs. V.A–V.D) is externally validated against wind-tunnel data; the dynamic-stability derivatives (Secs. V.E–V.F) are reported honestly as conservative engineering estimates with explicit disclosure of the closure adjustments they carry.

A. Body Normal Force and Center of Pressure

Subsonic body normal-force slope follows the Barrowman slender-body result $C_{N_\alpha,\,\mathrm{body}} = 2$. At supersonic speeds an Allen–Perkins crossflow term [32] augments the slender-body baseline,

$$C_{N_\alpha,\,\mathrm{body}}(M,\alpha) = 2 + K_{\mathrm{cf}}(M)\, C_{d,c}(M_c)\, \frac{A_{\mathrm{plan}}}{A_{\mathrm{ref}}}\, \frac{2\sin\alpha}{\pi}, \tag{22}$$

where K_{cf} is a crossflow multiplier that blends from the subsonic Galejs value to 1.1–1.3 supersonically, $C_{d,c}(M_c)$ is the Jorgensen crossflow drag coefficient [33] evaluated at the crossflow Mach $M_c = M\,|\sin\alpha|$, and $A_{\mathrm{plan}}/A_{\mathrm{ref}}$ is the ratio of planform to reference area. The asymptotic crossflow value $C_{d,c} = 1.20$ tabulated in NASA TR R-474 [33] is reproduced exactly. The body center of pressure migrates aft through the transonic band as the lift distribution transitions from the subsonic potential-flow pattern to the supersonic slender-body distribution; this shift is parametrized as a Mach-dependent fraction of body length over $M = 0.8$–1.3.

B. Fin Normal Force with Local Flow

The fin normal-force model retains the Barrowman three-coefficient decomposition $C_{N_\alpha} = K_1 + K_2\alpha + K_3\alpha^2$ but evaluates each coefficient at the *local* post-shock Mach M_s returned by the pre-pass rather than at freestream M_∞. Because the Prandtl–Glauert factor $\beta = \sqrt{M^2-1}$ is strongly nonlinear near unity, a 14% reduction in Mach at the fin station — representative of a 15° cone at $M_\infty = 2.5$ — translates into roughly an 18% change in $K_1 = 2/\beta$, so the use of local flow is the dominant correction in this regime. A Mach-dependent floor prevents K_1 from collapsing to numerical noise at high Mach:

$$K_{1,\,\mathrm{floor}}(M) = 0.85 - 0.45\big[1 - \exp\!\big(-K_{\mathrm{decay}}(M-1)\big)\big]\,, \qquad K_{\mathrm{decay}} = 1.480. \tag{23}$$

The decay constant was calibrated against the NASA TM X-653 wind-tunnel database [34] of finned-body normal-force slope and center-of-pressure measurements. Across $M = 0.6$–5.82 the calibrated model reproduced the measured C_{N_α} with a mean absolute percentage error (MAPE) of 6.84% and the center-of-pressure location x_{CP}/d with a MAPE of 7.11% over ten test points each (Fig. 15). The larger residual on x_{CP} is concentrated in the transonic band $M = 0.9$–1.2, where small absolute pressure shifts produce large percentage errors in the lever arm.

C. Pitts–Nielsen–Kaattari Interference

Fin–body and body–fin lift interference are handled by the Pitts–Nielsen–Kaattari (PNK) factors [35] K_{WB} and K_{BW}, generalized here to Mach-dependent form,

$$C_{N_\alpha,\,\mathrm{vehicle}} = K_{WB}(M_s)\, C^{\mathrm{alone}}_{N_\alpha,\,\mathrm{fin}} + K_{BW}(M_s)\, C_{N_\alpha,\,\mathrm{body}}, \tag{24}$$

with both factors evaluated at the local post-shock Mach M_s. A smoothstep blend over $M = 0.85$–1.15 activates the Mach-dependent correction from the subsonic limit $K = 1.0$. At $M = 2$–3 the corrections are typically 5–20% of the freestream-evaluated baseline. The interference treatment includes a floor on K_{WB} to preserve a physically admissible fin contribution at high Mach; the mechanism ablation of Sec. II.G shows that this floor has no measurable effect on integrated apogee on the validation corpus, consistent with its role as a local-loads correction rather than a trajectory driver.

D. ESDU Transonic Similarity

The near-sonic peak in fin C_{N_α}, where conventional thin-wing theory breaks down, is captured by a transonic similarity rule. The fin normal-force coefficient in the transonic band is mapped onto a universal function $h(K_{\text{trans}})$ of the transonic similarity parameter

$$K_{\text{trans}} = \frac{M^2 - 1}{[\tau\,\text{AR}]^{2/3}}, \qquad \tau = t/c, \tag{25}$$

where AR is the fin aspect ratio and τ the thickness-to-chord ratio. The universal curve is active for $K_{\text{trans}} \in [-2, +3]$ and is blended with the conventional $K_1/K_2/K_3$ formulation outside that band, restoring continuity with the supersonic local-flow model of Sec. V.B.

E. Pitch Damping (C_{m_q})

The pitch-damping derivative is computed by classical strip theory, summing the contributions of each aerodynamic component [36],

$$C_{m_q} = -2 \sum_{i=1}^{n_{\text{comp}}} C_{N_{\alpha,i}} \frac{(x_{CP,i} - x_{CG})^2}{L_{\text{ref}}^2}, \tag{26}$$

with the secondary $C_{m_{\dot\alpha}}$ derivative set to $0.4\,C_{m_q}$ following the Tobak–Wehrend slender-body theoretical ratio. A transonic Gaussian augmentation represents the unsteady shock-oscillation amplification of the damping near $M = 1$:

$$k_{\text{transonic}}(M) = 1 + 2.5\,\exp\left[-\left(\frac{M-1}{0.15}\right)^2\right], \qquad k(1.0) = 3.5, \tag{27}$$

which decays to unity within ±0.3 Mach of the transonic center (Fig. 17).

Two limitations of this derivative must be stated plainly. First, strip theory systematically overpredicts the C_{m_q} magnitude of an isolated axisymmetric body — by roughly a factor of five to ten relative to the exact slender-body theory of Tobak and Wehrend (NACA TN 3788) [36] (Fig. 16). The sign is correct and the resulting damping is conservative (over-damped, and therefore trajectory-safe), but the magnitude is not a quantitative result. Second, to close the trajectory on the validation corpus the production code applies a 3× multiplier on C_{m_q}. This multiplier is calibrated against the AEDC-TR-76-58 transonic pitch-damping wind-tunnel data; it has no independent flight-scale anchor and is disclosed as a *B-level* adjustment. Recent computational determinations of dynamic-stability derivatives indicate that, at $M = 1.08$–1.11, the augmented model overshoots the physical transonic pitch-damping peak resolved by the Sznajder CFD by roughly 110–160% [37]. The transonic Gaussian factor of Eq. (27) — whose peak augmentation parameter is $k(1.0) = 3.5$, a distinct model quantity from the overshoot just stated — is correspondingly only a qualitative match to free-oscillation behavior and is not claimed as a validated quantity. Both the multiplier and the Gaussian augmentation are expected to be replaced once modal C_{m_q} identification data for slender finned configurations become available.

F. Magnus, Vortex Sideforce, and Roll Damping

For a spinning rocket at angle of attack the Magnus side-force derivative is modeled as a fixed fraction of the body normal-force slope [38],

$$C_{y,\,p\alpha} = -\tfrac{2}{3}\,C_{N_\alpha,\,\mathrm{body}}, \tag{28}$$

where the factor $-\frac{2}{3}$ is the implemented Magnus side-force fraction applied to the body normal-force slope. In production the body contribution is not separated component-by-component; instead the body slope is approximated as a fixed fraction of the total vehicle slope, $C_{N_\alpha,\,\mathrm{body}} \approx 0.30\,C_{N_\alpha,\,\mathrm{total}}$, so that the effective Magnus fraction of the total slope is $-\frac{2}{3} \times 0.30 = -0.20$. The body-fraction value 0.30 and the Magnus fraction $-\frac{2}{3}$ are therefore distinct quantities, not alternative statements of the same coefficient. The corresponding Magnus yawing-moment derivative is $C_{n,\,p\alpha} = C_{y,\,p\alpha}\,(x_{CP} - x_{CG})/L_{\mathrm{ref}}$. At angles of attack above approximately 20° the asymmetric body vortex system generates a side force that is independent of roll angle. The present work models this contribution with a vortex-strength coefficient $K_v = 0.20$ and a ramp activation from $\alpha = 20°$ to 30°. We emphasize that $K_v = 0.20$ is an *internally calibrated* coefficient adopted for range-checking the high-incidence side force; it is not anchored to an external benchmark, and no literature value is claimed for it. Roll damping is computed analytically from the integrated fin chordwise pressure distribution under uniform roll rate, with corrections for fin–body interference applied through the PNK factors of Sec. V.C.

G. Shock–Boundary-Layer Interaction at Fin Roots

At $M > 1.2$ the fin leading-edge shock can separate the body boundary layer ahead of the fin, reducing the effective aerodynamic chord. The free-interaction theory of Chapman, Kuehn, and Larson [39] gives the separation-length scaling

$$\frac{L_{\mathrm{sep}}}{\delta} \propto (M^2 - 1)^{-1/4}, \tag{29}$$

clamped from below by $M^2 - 1 \geq 0.1$ to prevent divergence near the sonic line, where δ is the local boundary-layer thickness. The separation length is subtracted from the fin streamwise root chord when computing $K_1/K_2/K_3$. The associated pressure-drag contribution from the separated region was found to overestimate measured drag in the present finned-body fixtures and is disabled in production pending a re-derivation against higher-fidelity separated-flow data.

VI. Subsystem Benchmark Roll-Up

A. Verification and Validation Methodology

The model suite was verified and validated under the discipline of the AIAA Editorial Policy on Numerical and Experimental Accuracy [13], which applies to engineering prediction methods as it applies to computational fluid dynamics: code verification against analytical or tabulated solutions, error quantification with an explicit metric and stated regime, and an iterative-convergence demonstration wherever an iterative solver appears. We distinguish two classes of evidence. *Code verification* anchors a subsystem against an exact or tabulated reference—NACA Report 1135 shock and expansion tables [12], the Taylor–Maccoll cone solution [11, 17], and the Rayleigh pitot relation—where the acceptance criterion is agreement to near machine precision. *Physical validation* anchors a subsystem against published wind-tunnel, ballistic-range, free-flight, or CFD data—NACA RM A52H28 [22], NACA TN 3393 [26], NASA TM X-653 [34], and the DTIC AD0487365 cone-cylinder range tests [31]—where the acceptance criterion is a stated mean absolute percent error (MAPE) or mean absolute error (MAE) gate appropriate to a semi-empirical engineering method. Iterative solvers (the oblique-shock and Taylor–Maccoll root finders) converge to a 10^{-12} relative tolerance; integrated time-step convergence is reported in Sec. VIII ($|s| \approx 0.98\%$ per 10% contraction of the step).

To grade the evidence consistently we adopt a four-level rubric. A row is *A-level* when it is matched against published external or tabulated data with a quantitative acceptance criterion; *B-level* when it is a source-anchored analytical or flight-corpus closure that has not been isolated against a dedicated published component dataset; *C-level* for internal consistency or numerical-integrity checks; and *D-level* for a calibrated heuristic without external closure. Only A-level rows are claimed as validated in the sense of the AIAA policy; B-level rows are disclosed as such throughout.

B. Externally Benchmarked A-Level Subsystems

Twenty subsystems currently meet the A-level standard: an independent external benchmark exists, and the model passes a quantitative acceptance gate against it. They are collected in Table 5, with the primary reference, Mach range, and headline metric for each. The tabulated rows span $M = 0.6$ to $M = 10$ across the analytical, atmospheric, drag, and stability families; corroborating sources for each subsystem are documented in the per-benchmark validation memos under `paper/data/md/`. A single externally anchored *negative* benchmark (the NACA RM-10 high-fineness finned body) is retained outside the A-level count to bound and explicitly exclude a geometry family that lies outside the calibration envelope; it is discussed in Sec. X rather than presented as a validating result.

The analytical and atmospheric foundations verify to near machine precision. The exported speed-of-sound table agrees with the U.S. Standard Atmosphere 1976 [15] to a maximum error of 0.016%; the oblique-shock θ–β–M solver reproduces NACA Report 1135 to a maximum angle error of 0.021%; the Prandtl–Meyer expansion matches to a maximum absolute turning-angle error of 0.004°; and the Taylor–Maccoll cone solver returns the published cone-shock angle to a maximum relative error of 0.825% against a 1% gate. These are code-verification results: the agreement reflects correct implementation of an exact theory, not an empirical calibration.

The drag and stability families are physical validations and carry the larger, honestly reported residuals characteristic of semi-empirical engineering closures. Nose-and-body foredrag matches NACA RM A52H28 across five shapes to an aggregate MAE of 0.029 (gate 0.035). Turbulent base drag matches NACA TN 3393 to a MAPE of 15.9%; the laminar base-drag closure attributed to Chapman matches the corresponding TN 3393 laminar rows to 4.4%. Finned-vehicle total drag, exercised on the Army–Navy Basic Finner reference projectile against the ballistic-range data of Dupuis and Hathaway [40], achieves a MAPE of 11.8% over eight points spanning $M = 1.08$ to 4.30 (Fig. 18); four pointwise residuals exceed 14%, so the claim is an aggregate-gate pass rather than a uniform pointwise match. Static stability, validated against the wing-body-tail data of NASA TM X-653 [34] over $M = 0.6$ to 5.82, returns a normal-force-slope MAPE of 6.84% and a center-of-pressure (x_{CP}/d) MAPE of 7.11% over ten points each (Fig. 15). A hypersonic extension of the cone foredrag method, taken against the DTIC AD0487365 cone-cylinder range tests [31], is reported as an *exploratory* (B-level) result rather than a validated A-level subsystem; it is presented with its thin-cone limitation in Sec. VI.D and is not counted among the A-level rows above.

Table 5 Representative externally benchmarked A-level subsystems, with primary reference, Mach range, and headline metric. Analytical and atmospheric rows are code-verification results (near-exact agreement); drag and stability rows are physical validations against published wind-tunnel, ballistic-range, free-flight, or tabulated data, with the semi-empirical residual reported as a mean absolute percent error (MAPE) or mean absolute error (MAE).

#	Subsystem	Primary reference	Mach range	Headline metric
1	Speed of sound	U.S. Std. Atm. 1976 [15]	sea level–80 km	max error 0.016%
2	Sutherland viscosity	Incropera / NIST [16]	150–500 K	MAPE 0.54% vs. tables
3	Normal-shock relations	NACA Report 1135 [12]	$M = 1.5$–10	max error < 0.01%
4	Oblique-shock solver	NACA Report 1135 [12]	$M = 1.5$–10, $\theta = 5$–40°	max angle error 0.021%
5	Prandtl–Meyer expansion	NACA Report 1135 [12]	$M = 1.2$–10	max abs angle error 0.004°
6	Taylor–Maccoll cone flow	Cone tables [11, 17]	$M = 1$–10	shock angle 0.825% (gate 1%)
7	Rayleigh pitot $C_{p,\max}$	NACA Report 1135 [12]	$M = 1$–10, 15 pts	max error < 0.01%
8	ShockGeometry pre-pass state	NACA 1135 + Taylor–Maccoll	$M = 1.5$–5	cone surface Mach 0.00%
9	Nose/body foredrag, 5 shapes	NACA RM A52H28 [22]	$M = 1.5$–3, 5 shapes	aggregate MAE 0.029 (gate 0.035)
10	Base drag, turbulent	NACA TN 3393 [26]	$M = 2.73$–4.48	MAPE 15.9%
11	Base drag, laminar (Chapman)	NACA TN 3393 laminar [26]	$M = 2.73$–4.48	MAPE 4.4%

continued on next page

Table 5 (continued)

#	Subsystem	Primary reference	Mach range	Headline metric
12	Fin wave drag (DATCOM 4.1.5.1)	NACA TN 3650 [24]	$M = 1.1$–2.5, 12 pts	MAPE 21% (0.00% vs. Ackeret)
13	Compressible skin friction (Van Driest II)	Hopkins & Inouye [18]	$M = 1.5$–9	best-of-class transformation
14	Static stability C_{N_α}	NASA TM X-653 [34]	$M = 0.6$–5.82, 10 pts	MAPE 6.84%
15	Static stability x_{CP}/d	NASA TM X-653 [34]	$M = 0.6$–5.82, 10 pts	MAPE 7.11%
16	Body-alone pitch damping C_{m_q} (no fins)	Tobak NACA TN 3788 [36]	$M = 1.5$–3	39% at $M = 1.5$; sign-correct, conservative (distinct from the finned Basic Finner C_{m_q}, Sec. VI.D)
17	Crossflow body $C_{d,c}$	Jorgensen TR R-474 [33]	supersonic crossflow	exact match (1.20)
18	Crossflow fin $C_{d,c}$	Hoerner / Jorgensen [30]	crossflow	1.42 (within range)
19	Magnus body fraction	Platou 1965 [38]	supersonic	0.30 within 0.30–0.80
20	Finned-vehicle total drag	Basic Finner [40]	$M = 1.08$–4.30, 8 pts	MAPE 11.8%

C. Secondary and Internally Calibrated Rows

Two rows that earlier internal documentation listed among the precision A-level benchmarks are presented here at their honest evidentiary level rather than as quantitative validations.

The AGARD Model B subsonic-to-transonic drag comparison against AEDC-TR-70-100 [41] is retained only as a *qualitative* secondary trend benchmark. The model reproduces the subsonic-to-transonic drag-rise trend over $M = 0.2$ to 1.0, but its total-drag MAPE on this configuration is 22.6%—a loose qualitative closure, not a precision match—and it is excluded from the A-level count and from any quantitative claim. The AGARD-B configuration is used in Sec. VII as a published-CFD reference geometry, where the same qualitative-trend framing applies.

The vortex-induced sideforce coefficient is governed by an internal ramp constant $K_v = 0.20$ that activates above roughly $\alpha = 20°$. No verifiable external benchmark could be located for this coefficient, so it is presented as an *internally calibrated* parameter constrained by an order-of-magnitude range check rather than anchored to a published dataset; the earlier literature-anchored A-level claim is withdrawn. It is reported for transparency and so that downstream users may re-calibrate it as suitable high-angle-of-attack data become available.

D. B-Level Disclosures and In-Sample Calibration

Several model decisions do not meet the A-level standard and are disclosed here.

The pitch-damping derivative C_{m_q} is augmented in production by a fixed 3× multiplier and a transonic Gaussian peak ($k = 3.5$ at $M \approx 1$), selected to close the angular-rate residual on the flight corpus. Three independent reference datasets quantify the consequence, and because they are distinct data they are reported separately rather than reconciled to a single number. (i) Against the Army–Navy Basic Finner *finned*-vehicle free-flight C_{m_q} data of Dupuis and Hathaway [40]—a different configuration and reference from the body-alone Tobak TN 3788 row (#16, 39% at $M = 1.5$) in Table 5—the augmented model is sign-correct but exhibits a supersonic MAPE near 69% (under-prediction) over the supersonic points. (ii) Against the independent Bhagwandin and Sahu URANS free-flight comparison [42] the same model returns a supersonic MAPE of 18.96% on the AFF planform (5 points, $M = 1.30$–2.50) and 28.02% on the ANF planform (8 points, $M = 1.29$–4.50), sign-consistent with the Basic Finner under-prediction. (iii) Against the engineering-model damping CFD of Sznajder [37], the transonic Gaussian over-predicts the free-oscillation damping peak by roughly +110 to +160% at $M = 1.08$–1.11; this overshoot is a comparison versus a different reference dataset and should not be conflated with the model's internal Gaussian augmentation parameter ($k = 3.5$ at $M = 1$) or with the $\sim 69\%$ Basic Finner free-flight figure. All three are reported as B-level so downstream users can re-identify the damping closure as further independent datasets become available.

The hypersonic cone foredrag benchmark, taken against the DTIC AD0487365 cone-cylinder range tests [31] with the source Reynolds number matched row by row, returns a MAPE of 19.7% over eleven points spanning $M = 6.5$ to 17.2, with the largest single residual reaching +57% on the thinnest, lowest-Reynolds-number cone (Fig. 14). The large residuals fall on the thinnest ($\theta_c \leq 8°$) cones at the lowest-Reynolds-number rows, where friction and base drag dominate and the AD0487365 boundary-layer transition state is incompletely specified. This row is therefore an *exploratory* extension of the supersonic-validated foredrag method and is reported as B-level rather than as a validated A-level subsystem; until a thin-cone dataset with a documented transition state is located, the hypersonic regime as a whole is treated as an exploratory extension rather than a validated operating range.

Finally, and most importantly for the integrated-flight claim, the finned-vehicle base-drag augmentation depends on two scale constants that are *corpus-frozen* rather than externally anchored: a thick-boundary-layer scale `THICK_BL_K` = 2.2 and a slender-body scale `SLENDER_BODY_K` = 0.0025. These were fixed against a small set of amateur flights—the anchor flights Raven, Rabia, Kinsel, and Torrent—so the flight-corpus apogee result of Sec. VIII is, to that extent, partly in-sample. We disclose this explicitly and do not present the corpus aggregate as a fully blind result. The principal defense against an overfitting interpretation is the decontaminated prospective holdout of Sec. VIII, in which every flight that either constant touched is placed in the development split: the genuinely blind holdout is *more* accurate than the development split, indicating that the two constants generalize rather than overfit. The reader is referred to Sec. VIII for the split construction and the corresponding error statistics.

VII. Validation Against Published Computational Fluid Dynamics

No in-house computational fluid dynamics (CFD) campaign was undertaken; this is a deliberate scoping decision, revisited in the limitations discussion. In its place the present method is compared against four primary published

CFD comparators—with a fifth study used as an independent corroborating source—together spanning two reference geometries, two distinct aerodynamic quantities, and three Mach bands. The four primary comparators are: the unsteady Reynolds-averaged Navier–Stokes (URANS) study of the Army–Navy Basic Finner by Bunescu et al. [43]; the thin-layer Navier–Stokes computation of a secant-ogive-cylinder-boattail projectile by Sahu, Nietubicz, and Steger [44]; the Menter shear-stress-transport (SST) k–ω solution of the AGARD Model B calibration standard by Vidanović et al. [45]; and the ANSYS Fluent pitch-damping computations of the Basic Finner by Sznajder [37]. A second independent CFD source on finned-projectile pitch damping, the CFD++ study of Bhagwandin and Sahu [42], is used to corroborate the Sznajder finding; that study reports the Army–Navy Finner (ANF, identical to the Basic Finner) and the geometrically distinct Air Force Modified Finner (AFF; tangent-ogive nose with clipped-delta fins), so it doubles as a second-planform cross-check rather than a same-geometry repeat. The inventory is summarized in Table 6, and the CFD-side panels are collected in Fig. 19.

A. Basic Finner Static Axial Force versus Bunescu URANS

Bunescu et al. [43] reported URANS predictions of normal- and axial-force coefficients on the standard four-rectangular-fin Army–Navy Basic Finner at $M = 0.4$, 0.95, 1.6, 2.5, and 3.5, using a realizable k–ε closure with corroborating SST k–ω comparisons. The present method was exercised on the identical geometry at the same five Mach numbers and at zero angle of attack. Absolute agreement in the axial-force coefficient C_X is loose: the largest residuals occur near the transonic point $M = 0.95$, where the shock-induced separation resolved by the URANS field is not represented by the engineering pressure-drag model, and at the supersonic points, where the semi-empirical viscous treatment differs from the Reynolds-number-resolved boundary layer. The Mach trend, however, is reproduced correctly: a transonic rise to a peak near $M \approx 1$ followed by monotone decay through $M = 3.5$. This comparison is reported as a qualitative trend check rather than as a quantitative benchmark.

B. Ogive-Cylinder-Boattail Base Drag versus Sahu Thin-Layer Navier–Stokes

Sahu, Nietubicz, and Steger [44] computed base, pressure, friction, and total drag for a secant-ogive-cylinder-boattail projectile across the transonic band $M = 0.9$–1.2 at $\mathrm{Re}_L = 4.5 \times 10^6$, overlaying their thin-layer Navier–Stokes solution against sting-mounted wind-tunnel data and a semi-empirical correlation. This triple comparison would provide an ideal anchor for a transonic base-drag panel against the present method's base-drag and pressure-drag pipelines. The digitization of this dataset was deferred during preparation of the present manuscript; the comparison is therefore retained as near-term future work rather than presented as a result, and the corresponding panel in Fig. 19 is flagged accordingly.

C. AGARD Model B Reference versus Vidanović SST k–ω

Vidanović et al. [45] reported ANSYS Fluent SST k–ω predictions of total drag, lift, and pitching-moment coefficients on the AGARD Model B calibration standard at $M = 0.596$ and $M = 1.602$ over an angle-of-attack sweep, validated against wind-tunnel data from the VTI T-38 trisonic facility. The published CFD-versus-experiment agreement is within a few percent in C_D and below one percent in C_L over the test envelope. The present method does not ship an AGARD-B build—the equilateral-triangle delta wing with a biconvex section lies at the edge of the validity of the underlying fin-set model—so no quantitative comparison is reported here. Consistent with the benchmark roll-up, the AGARD-B configuration is treated as a qualitative trend reference only (the corresponding total-drag agreement is loose, with mean absolute percentage error of order 22%), and the Vidanović CFD is displayed in Fig. 19 as a reference dataset against which a future comparator build can be benchmarked. The omission is acknowledged as a deferred future-work item.

D. Basic Finner Pitch Damping versus Sznajder Fluent, Corroborated by Bhagwandin and Sahu

The most informative CFD comparison concerns the pitch-damping derivative on the Basic Finner. Sznajder [37] computed C_{m_q} and $C_{m_{\dot{\alpha}}}$ using three independent CFD techniques—a steady moving-reference-frame method, a dynamic-mesh forced-oscillation method, and a step-perturbation indicial-response method—over $M = 0.9$–5.0. The three methods agreed to within approximately 3% of one another and were independently validated against free-flight experimental data. The present method exposes the experimentally observable damping sum $C_{m_q} + C_{m_{\dot{\alpha}}}$, which was compared against the Sznajder solution on a common grid with the following two findings.

- *Supersonic band*, $M = 1.29$–4.5 ($n = 8$ points): the present method underpredicts the magnitude of the damping sum by 27 to 36%, with the sign and Mach trend correct. The mean absolute percentage error over this band is 31.6%.
- *Transonic peak*, $M \approx 1.08$–1.11 ($n = 2$ points): the present method overshoots the magnitude of the damping sum by 110 to 160%; the CFD does not exhibit a comparable peak in the sum.

The transonic overshoot is traced to the Gaussian augmentation applied to the strip-theory damping near $M = 1$ in the stability model; the supersonic underprediction reflects a constant-factor bias in the strip-theory damping coefficient. Bhagwandin and Sahu [42] provide an independent CFD++ pitch-damping reference on two finner planforms—the ANF (identical to the Basic Finner) and the geometrically distinct AFF—validated against the same free-flight family. Restricting attention to the supersonic band where their CFD itself converges with the free-flight scatter, their comparator against the present method yields a mean absolute percentage error of 19.0% on the AFF ($n = 5$, $M = 1.30$–2.50) and 28.0% on the ANF ($n = 8$, $M = 1.29$–4.50), with the same sign and the same direction of the residual on every supersonic point. The full-envelope figure is larger because it is dominated by the same transonic Gaussian over-augmentation seen against Sznajder, not by an independent defect.

The two CFD sources therefore converge on the same two findings: a supersonic underprediction of pitch damping with correct sign and trend—27–36% against Sznajder, 19–28% against Bhagwandin and Sahu across the two planforms—and a transonic-peak over-augmentation. Because the second source is on the AFF and ANF finner fixtures (the AFF planform is reconstructed from the source figure and is itself a B-level fixture) rather than an exact replicate of the Sznajder build, this is a cross-planform corroboration of the residual's sign and trend rather than a point-for-point confirmation of magnitude. Consistent with this evidence, the transonic pitch-damping augmentation is held to be a B-level model element rather than a validated one. Both findings are taken up explicitly in the limitations discussion.

VIII. Flight-Corpus Integration Test

The component benchmarks of the preceding sections isolate individual physical models against canonical wind-tunnel and range data. The integration test asks the question that matters operationally: when every model is composed inside a full six-degree-of-freedom trajectory and run end to end against real flights, how accurately does the predictor recover the measured apogee? This section reports that result on an externally defined corpus, quantifies its uncertainty with non-parametric inference, and discloses the in-sample dependence honestly.

A. Corpus Definition and Inclusion Criterion

The headline corpus comprises 25 flights spanning peak Mach 0.54 to 4.33: 23 single-stage vehicles together with 2 two-stage flights (the AeroPac 104K flight at Mach 3.04 and the MESOS 293K flight at Mach 4.33), drawn directly from the public RASAero II flight-comparison set assembled by Rogers [4, 46]. The MESOS 293K flight closes the corpus at its highest Mach. The inclusion criterion is the central guard against selection bias: every flight is taken from a comparison set chosen and published by an *external* party for an independent purpose, and the set is admitted in its entirety. No flight was added, removed, or reweighted on the basis of how well the present predictor happened to perform on it. The accuracy statistics below are therefore an outcome-independent validation result rather than a curated demonstration. Vehicles range from 1.75 in hobby airframes to the 8.0 in Qu8k and the 6.13 in A-601 Kinsel, and ground-truth apogees derive from heterogeneous instrumentation (barometric altimetry, optical tracking, integrated accelerometry, and GPS), as recorded in Table 9.

A separate set of high-Mach sounding rockets reaching Mach 7 is treated as an *exploratory* study in Sec. IX, deliberately segregated from this headline validation; the present corpus is supersonic-validated and contains no hypersonic flight (maximum Mach 4.33).

B. Headline Accuracy

Across the 25 flights, the OpenRocket-Plus apogee-prediction error (ORP, defined as $100\,(h_{\text{pred}} - h_{\text{meas}})/h_{\text{meas}}$) has a mean signed value of -0.38 % with sample standard deviation $\sigma = 5.44$ %, root-mean-square error of 5.34 %, and mean absolute error of 4.74 %. All 25 of 25 flights fall within ±10 %, and 14 of 25 within ±5 %; the largest single residual is 8.7 %. Table 7 summarizes these figures alongside the recorded RASAero II predictions for the same flights.

Because the corpus deviates from normality (Sec. VIII.D), every interval below is a 95% non-parametric percentile bootstrap confidence interval (20 000 resamples, fixed seed `0x51A7EA`). The mean-error interval is $[-2.41, +1.72]$ %; it brackets zero, so no bias is detectable on this corpus. We are careful not to overstate this as proven unbiasedness: at $n = 25$ the interval is roughly ±2 pp wide, so a small true bias would not be resolved, and bracketing zero is an absence of evidence for bias rather than evidence of its absence. The remaining intervals are $[4.20, 6.25]$ % for σ, $[4.38, 6.19]$ % for RMSE, and $[3.78, 5.70]$ % for MAE. The within-±5 % rate of 56 % carries an interval of $[36, 76]$ %. The within-±10 % rate is 100 % (25 of 25); because the bootstrap is degenerate for an all-pass proportion (every resample contains only passing flights, returning $[100, 100]$ as an artifact), we instead report an exact binomial Clopper–Pearson interval, whose lower bound is 86.3 % at 95 % two-sided confidence. Decomposing the mean squared error into squared bias and variance yields a bias^2/MSE ratio of 0.01 for ORP, against 0.16 for RASAero II: the ORP residual is essentially pure variance. Operationally this means the remaining error is dominated by irreducible flight-to-flight scatter—build tolerance, motor lot variation, and atmosphere—rather than a directional defect in the aerodynamic model. Figure 21 visualizes this decomposition and Fig. 20 shows the residuals against peak Mach.

C. Per-Regime Behavior

Partitioning the corpus by Mach regime (Table 8) localizes the error structure. The subsonic ($M < 0.8$, $n = 9$) and low-supersonic ($1.3 \leq M < 3.0$, $n = 5$) bands carry small positive biases (2.54 % and 1.83 %). The transonic band ($0.8 \leq M < 1.3$, $n = 7$) is the disclosed regime weakness, with a mean bias of −3.66 %: this is the Mach range where shock-induced drag rise is most sensitive to geometry detail and where the present models are weakest. The high-supersonic band ($3.0 \leq M < 5.0$, $n = 4$) has a −3.98 % bias but the tightest scatter ($\sigma = 2.97$ %). Above Mach 3 the sample is small ($n \leq 4$ per band), so these figures are reported descriptively; no inferential claim is made there. The hypersonic regime ($M > 5.0$) is empty by construction. The regime breakdown is shown graphically in Fig. 22.

D. Distribution and Choice of Inference

The choice of non-parametric inference is dictated by the residual distribution, not adopted by convenience. The Shapiro–Wilk test rejects normality of the 25 ORP residuals ($W = 0.905$, $p = 0.023$ at $\alpha = 0.05$), and the Anderson–Darling statistic ($A^2 = 0.922$) likewise exceeds its critical value (0.728). The residuals are mildly right-skewed (skewness +0.43) and distinctly platykurtic (excess kurtosis −1.14): the distribution is light-tailed rather than heavy-tailed, consistent with the bounded maximum residual of 8.7 %. We therefore adopt the Wilcoxon signed-rank test as the primary hypothesis test and bootstrap confidence intervals in place of normal-theory intervals throughout. We also caution that, at $n = 25$, normality tests have limited power; failing to reject normality would not establish it, which is a further reason to prefer distribution-free methods. Figure 23 compares the ORP and RASAero II error distributions, and Fig. 24 shows the bootstrap sampling distributions underlying the headline intervals.

E. Paired Comparison Against RASAero II

Because each flight carries a recorded RASAero II prediction, the two methods can be compared head to head on identical vehicles and motors. On absolute error, the paired mean difference $|\mathrm{ORP}| - |\mathrm{RAS}|$ is -0.60 pp with a 95% non-parametric percentile bootstrap confidence interval of $[-2.16, +0.96]$ pp, which straddles zero. The Wilcoxon signed-rank test on the paired absolute errors returns $W = 143.0$, $p = 0.615$: no significant difference in accuracy is detected on this corpus, with ORP closer on 13 flights and RASAero II closer on 12. We read this as a failure to reject rather than as positive proof of equivalence: at $n = 25$ the paired test has limited power, so $p = 0.615$ alone is absence of evidence for a difference, not evidence of its absence. Equivalence is supported more defensibly by the bootstrap interval on the paired mean above, which lies entirely within roughly ± 2 pp—a two-one-sided-test argument against any difference larger than that margin—than by the non-significant Wilcoxon p-value. The Bland–Altman limits of agreement are ± 14.3 % about a mean offset of -2.84 %. The honest claim supported by these statistics is *parity*, not superiority. We further disclose an important asymmetry: the RASAero II values are Rogers' published recorded predictions for these flights, not fresh pre-flight reruns we generated and can independently verify, so the comparison is to a version-locked external record rather than a controlled re-execution. In addition, for the two two-stage flights—AeroPac 104K and MESOS 293K—the recorded RASAero II apogees are *post-flight* simulations in which the ignition delay and launch angle were adjusted to match the measured flight; the head-to-head on these two flights is therefore not a blind forward prediction, and the corresponding RASAero II errors should be read accordingly. Figure 25 shows the paired residuals.

F. In-Sample Dependence and Decontaminated Holdout

Two base-drag scale constants are calibrated at the corpus level and are therefore not free of in-sample dependence: `THICK_BL_K` = 2.2 (anchored to the Raven flight) and `SLENDER_BODY_K` = 0.0025 (anchored to Raven, Rabia, and Kinsel, with Torrent used as a source diagnostic). The headline statistics are thus partly in-sample, and we report this rather than obscure it.

The primary defense against overfitting is a decontaminated prospective holdout. Every flight that either of the two constants touched—Raven, Rabia, Rabia Short Fin Can, Kinsel, and Torrent—is placed in the development partition, leaving a genuinely blind holdout. On the development partition ($n = 13$) the mean error is 0.22 % with MAE 5.47 %; on the blind holdout ($n = 12$) the mean error is -1.03 % with MAE 3.95 %. The holdout is *more* accurate than the development set, which is the signature of constants that generalize rather than overfit. (An earlier split left Rabia in the holdout, contaminating it; the decontaminated split corrects this.) This prospective generalization, not any operational sweep, is the basis on which we rest the in-sample disclosure.

G. Two-Stage Closure and a Disclosed Regression

The MESOS 293K flight closes the corpus at Mach 4.33 and is one of the two two-stage flights in the corpus (alongside AeroPac 104K), exercising staging and coast-phase modeling not present in the single-stage flights. We disclose a regression here relative to the RFD v1.2 archived snapshot: that earlier code version recorded an apogee of 291 601 ft (−0.6 %), whereas the current archived code release predicts 273 056 ft, an error of −6.96 %. The −6.96 % value is reproducible from the current archived code release—verified by running the MESOS flight in isolation—and the root cause of the regression is under investigation; the next RFD release will re-sync the model column. The flight remains within the ±10 % envelope and is retained in the corpus under the same outcome-independent inclusion rule applied to every other flight; we do not exclude it on the basis of the degraded result. The regression is carried forward to the limitations discussion.

H. Per-Flight Results

Table 9 lists all 25 flights with vehicle, motor, peak Mach, measured and predicted apogees, and per-predictor error, ordered by peak Mach.

IX. Exploratory High-Mach Sounding-Rocket Assessment

The 25-flight headline corpus of Sec. VIII is bounded above by Mach 4.33, so it contains no hypersonic flights. To probe whether the pre-pass and its downstream supersonic models continue to function beyond that ceiling, we assembled a separate set of roughly twenty historical sounding-rocket flights with documented apogees, several of which peak between Mach 5 and Mach 7. This set is *not* part of the validation corpus and is reported here only as an exploratory probe of method reach. We state at the outset that these flights are *excluded from every headline statistic*: presenting only the three that happen to fall within ±10% would constitute outcome-based selection bias, the single most serious threat to the credibility of the result. We therefore report the full set—successes and failures alike—in Table 10.

A. Full Exploratory Set

Of the historical flights simulated, three fall within ±10% of the measured apogee. The Black Brant V VB (AAF-VB-32, Churchill Research Range, 3 March 1971; [47]) reaches a peak Mach number of 7.22 and is predicted to within −6.97% of its 273.6 km measured apogee. The two Nike–Deacon (DAN) meteorological flights of [48], peaking at Mach 4.96 and Mach 5.08, are predicted to −1.06% and −0.89% respectively. These three results establish that the integrated method *reaches* Mach 7 and returns apogee predictions within about 7% *when the vehicle is well characterized*—that is, when an experimentally documented thrust history and a complete external geometry are available, as they are for these flights.

The remaining flights fall well outside ±10%, and they fail in two opposite directions. The 1965 Nike–Apache set, whose vehicle definition follows the NASA performance handbook [49] and whose measured apogees are taken from the NASA Wallops range report X-721-67-103, is over-predicted by roughly +24% to +36%, and the Nike–Cajun sounding flight (NACA RM L57D26) is over-predicted by +16.6%. The original blunt and secant-ogive Arcas flights (DTIC AD-235341) and the HEROS 3 amateur flight (Kobald, 2018) are instead severely under-predicted, by −29% to −69%. Two cases are outright simulation failures rather than honest mispredictions: Nike–Apache 14.210 GI returns a degenerate zero apogee (a launch-state geometry/recovery configuration error), and the Nike–Cajun Hurricane flight terminates with an integration error before apogee; both are listed in Table 10 for completeness but carry no physical interpretation.

B. Root-Cause Assessment

The dominant source of error in this set is *vehicle reconstruction*, not a single defect in the aerodynamic models. The headline corpus of Sec. VIII pairs each flight with a manufacturer or contest-documented thrust curve and a complete airframe definition; the historical sounding rockets here do not. Their thrust histories were synthesized from total-impulse and burn-time figures, and their external geometries were reconstructed from partial drawings and performance handbooks. The two opposite-signed failure modes are consistent with this diagnosis rather than with a common modelling bias. The Arcas and HEROS 3 under-predictions, which occur at modest peak Mach numbers (1.9–2.5), are the signature of under-stated propulsive energy in the reconstructed motors—an 178,000 ft documented Arcas apogee cannot be matched by any drag model when the synthesized impulse is too low, and the error there is propulsive bookkeeping, not high-Mach aerodynamics.

The Nike–Apache over-predictions are more interesting because they are *systematic and same-signed* across nine independent flights (+24% to +36%), all peaking above Mach 5.9. A purely random reconstruction error would not bias an entire vehicle family in one direction. This pattern instead points to an aerodynamic or flight-dynamics mismatch specific to this configuration above Mach 5—most plausibly an under-predicted coast-phase drag coefficient (too little aerodynamic deceleration during the long high-altitude coast to apogee), together with the simplified thrust-misalignment and dispersion assumptions of a nominal vertical trajectory. Identifying which of these dominates would require the detailed range and trajectory data that motivate treating the Nike–Apache result as an open question rather than a validated outcome.

C. Status and Future Work

We draw no validation claim from this set. The honest reading is that the method's supersonic models continue to run, and can be accurate to within about 7% at Mach 7, but only on the small number of flights for which the propulsion and geometry are independently documented to the standard of the headline corpus. On poorly documented historical vehicles, reconstruction uncertainty swamps any model signal. Establishing genuine hypersonic flight validation therefore requires either flights with experimentally measured thrust and mass-property histories at these Mach numbers, or a dedicated reconstruction-uncertainty quantification that propagates the documented tolerances on impulse, mass, and geometry into the predicted apogee. Both are left to future work; until then the paper's validated envelope remains the supersonic regime to Mach 4.33 established in Sec. VIII, and the high-Mach reach demonstrated here is reported as exploratory only.

X. Limitations and Honest Disclosures

The framework is validated across the component benchmarks of Sec. VI and the externally selected 25-flight corpus of Sec. VIII, but several limitations bound the scope of admissible inference. Each is disclosed below with its quantitative magnitude, its root cause where identified, and either an in-paper mitigation or a documented future fix. We state these plainly so that the headline claim—supersonic parity with RASAero II to $M = 4.33$, with hypersonic reach demonstrated only as an exploratory method capability—is not over-read.

A. Transonic regime weakness

The largest disclosed model weakness is in the transonic band. Across the $n = 7$ headline flights with peak $M = 0.8$–1.3, OpenRocket-Plus carries a mean signed apogee error of -3.66% ($\sigma = 5.36\%$, RMSE 6.17%, MAE 5.84%), the only regime in which RASAero II [4] is the more accurate predictor. The likely cause is that the supersonic-tuned blending of the wave-drag and base-drag stacks pulls total drag slightly low on the subsonic side of $M = 1$, over-predicting apogee. The transonic pitch-damping augmentation provides an independent corroboration of the same calibration gap: the Gaussian C_{mq} multiplier overshoots the transonic damping peak by +110 to +160% at $M = 1.08$–1.11 relative to the Sznajder Fluent reference [37]. Two independent observations—flight-corpus apogee bias and CFD-benchmarked C_{mq}—therefore converge on the transonic blending region as the principal open calibration gap. The proposed fix is to redistribute transonic component wave drag through a Whitcomb equivalent body of revolution rather than summing the components independently.

B. Partly in-sample headline (two B-level base-drag constants)

The headline corpus is *partly* in-sample. Two base-drag scale constants are B-level and corpus-frozen: a thick-boundary-layer factor `THICK_BL_K` = 2.2 (anchored on a single flight, Raven) and a slender-body factor `SLENDER_BODY_K` = 0.0025 (anchored on Raven, Rabia, and Kinsel, with Torrent used in the source diagnostic). Because these constants were tuned against flights that also appear in the headline statistics, the headline cannot be read as fully blind. The mitigation is a *decontaminated* prospective holdout in which every flight any constant touched (Raven, Rabia, Rabia Short Fin Can, Kinsel, and Torrent) is placed in the development partition. On that split the development partition ($n = 13$) gives a mean error of +0.22% and MAE 5.47%, while the genuinely blind holdout ($n = 12$) gives a mean of −1.03% and MAE 3.95%. The holdout is *more* accurate than the development set, which is the primary evidence that the two constants generalize rather than overfit. We note that an earlier split inadvertently placed a constant-touched flight (Rabia) in the holdout; the decontaminated split reported here corrects that contamination.

C. Sparse high-Mach sampling in the headline corpus

The headline corpus peaks at $M = 4.33$; its hypersonic regime ($M > 5$) is *empty*. Of the 25 flights, 9 are subsonic ($M < 0.8$), 7 transonic, 5 low supersonic ($M = 1.3$–3.0), and only 4 high supersonic ($M = 3.0$–5.0). Above $M = 3$ the per-regime statistics are descriptive only ($n \leq 5$); no inferential claim is attached to them. Consequently the hypersonic capability is reported in Sec. IX strictly as an *exploratory* result—the method *reaches* $M = 7$ within ±7% on well-characterized vehicles (Black Brant VB at $M = 7.22$, −6.97%; two Nike-Deacon flights at $M \approx 5$, −1.06% and −0.89%)—but on poorly documented historical flights the motor- and geometry-reconstruction uncertainty dominates, and the method is *not* validated in the hypersonic regime. We deliberately do not present the high-Mach set as a within-tolerance headline, which would constitute outcome-based selection.

D. Two-stage closure (MESOS 293K) regression

The corpus contains two two-stage closures (the AeroPac 104K flight at $M = 3.04$ and MESOS 293K at $M = 4.33$); the remaining 23 flights are single-stage. MESOS 293K currently predicts a signed apogee error of −6.96% (273,056 ft). This is a *disclosed regression* relative to the archived RFD v1.2 snapshot, which recorded 291,601 ft (−0.6%) under an earlier code version. The −6.96% value is itself reproducible: it is reproduced from the current archived code release with the MESOS flight run in isolation. The root cause of the regression from the v1.2-era value is under investigation; we do *not* attribute it to a specific mechanism here. The current −6.96% result remains within the ±10% admission criterion and so does not change the 25/25 headline. The next RFD release will re-synchronize the model column to the current code; we report the reproducible −6.96% value throughout this paper.

E. No author-produced computational fluid dynamics

The validation in Sec. VII cites four primary published-CFD comparators (with a corroborating fifth source) rather than runs produced by the present author; this is a deliberate scoping decision for a solo-author, self-funded open-source contribution. The mitigation is breadth: the comparators span two geometries (Basic Finner and AGARD-B), two coefficient families (static drag and dynamic pitch damping), the transonic through hypersonic-limit Mach bands, and four independent author groups—Bunescu URANS [43], the Sahu and Bhagwandin TLNS and follow-up work [42, 44], Vidanović SST k–ω [45], and Sznajder Fluent [37]. Future work is to ship a closed-loop comparator pipeline—an author-built AGARD-B model driving the Vidanović reference and an ogive-cylinder-boattail model driving the Sahu base-drag reference—to remove the "reference dataset only" caveat.

F. B-level transonic pitch-damping multiplier

The transonic pitch-damping augmentation is held at B-level confidence. The strip-theory damping model applies a single transonic multiplier of approximately 3×, calibrated against AEDC-TR-76-58 wind-tunnel data [50] and re-checked against the Sznajder Fluent reference [37], against which it overshoots the transonic damping peak by +110 to +160% at $M = 1.08$–1.11 (Sec. X.A). The root cause is the absence of a flight-scale anchor: the multiplier rests on this single wind-tunnel-scale source with no independent flight-scale validation. This affects predicted dynamic stability and coning behavior but not the apogee statistics, which are insensitive to C_{mq}. Lifting this term to A-level requires an independent flight-scale pitch-damping dataset, which is identified as future work.

G. Heterogeneous ground-truth sources

The corpus ground truth is heterogeneous, drawn from GPS- and barometric-altimeter apogee records, optical and radar tracking, and published flight reports of differing vintage and instrumentation. Reported apogee precisions across these sources imply an irreducible measurement floor on the order of 1% that is *not* subtracted from the reported errors. The headline mean signed error of −0.38% and RMSE of 5.34% therefore include this measurement contribution; the true model-only error is correspondingly smaller, but we make no attempt to deconvolve it and report the raw, conservative figures throughout.

H. Aeroelastic model implemented but disabled

A fin aeroelastic-effectiveness model is present in the codebase but is gated by a dynamic-pressure threshold of $Q_{\text{THRESHOLD}} = 1 \times 10^{12}$ Pa and is therefore effectively disabled in every simulation reported here. No aeroelastic claims—flutter, divergence, or fin-tip-twist-driven $C_{N\alpha}$ reduction—are made in this paper. The threshold is held at 10^{12} Pa pending an author flutter-and-divergence validation campaign against published cantilever-fin data, after which the gate can be lowered to a physically meaningful value.

I. Sensitivity sweep run on the exploratory high-Mach vehicles

The four-flight operational sensitivity sweep reported below was run on the high-Mach *exploratory* vehicles (HEROS 3, the blunt Arcas, a Nike-Apache, and Black Brant VB) rather than on the gentler headline flights. These are the fastest, highest-Mach trajectories in the dataset and therefore bound the numerical-convergence behavior of the milder headline cases as a worst-case envelope: drag-coefficient scale dominates the response (mean local sensitivity $|s| \approx 4\%$ apogee per 10% C_d, i.e. a sensitivity coefficient near 0.4), while the integration time step contributes $|s| \approx 0.98\%$ over the range 0.025–0.10 s, confirming numerical convergence per the AIAA accuracy policy [13]. We state explicitly that the sweep vehicles are exploratory: the in-sample defense of the headline rests on the decontaminated holdout of

Sec. X.B, not on this sweep.

XI. Conclusions and Future Work

A. Conclusions

This paper has extended an open-source rocket trajectory simulator from a subsonically valid Barrowman baseline [1, 3] to a compressible framework whose integrated apogee accuracy is quantified, and benchmarked against an independent semi-empirical predictor, across the supersonic regime to Mach 4.33. The work makes three contributions, framed here as honestly as the supporting evidence permits.

First, we introduced a *shock-geometry pre-pass* architecture. Once per timestep the pre-pass walks the airframe from nose to tail and distributes locally corrected post-shock Mach number, pressure, and temperature to every downstream drag, normal-force, and stability calculator. The pre-pass is inert below Mach 1, so it adds no cost to subsonic flight, and its surface-flow solution is verified bit-for-bit against the classical references: cone surface conditions agree with Taylor–Maccoll to within a 0.825% shock-angle error against the gate of 1% [11], and shoulder expansions reproduce Prandtl–Meyer turning angles to within 0.004° [12]. We are deliberate about what this buys. A mechanism ablation (Sec. II.G) shows the pre-pass moves *integrated apogee* by only 0.15 percentage points in the mean, because apogee integrates a trajectory dominated by lower-Mach drag and the pre-pass is inert below Mach 1. Its value is therefore *local-flow fidelity*—correct post-shock conditions for fin loads and stability—and its role as the architectural seam that makes the downstream supersonic models physically consistent, not a gross-apogee improvement. The dominant apogee mechanism in that same ablation is the externally calibrated finned-base drag augmentation (8.10 percentage points mean effect). The dominance of this finned-base augmentation in the apogee error budget is examined in detail—as a distinct research question, not a re-slice of the present integrated result—in a companion base-drag intercomparison study [14].

Second, we replaced the underlying engineering models with externally benchmarked subsystems (Secs. IV–VI), each verified against published wind-tunnel, free-flight range, or computational data with a documented error metric. These include Van Driest II compressible skin friction [18], the DATCOM fin wave-drag method [23], a supersonic base-drag stack validated against NACA TN 3393 [26] and consistent with ESDU 77021 [25], and Modified Newtonian impact pressure [17]. Representative component accuracies are total drag on the Basic Finner range projectile at MAPE 11.8% (eight points, Mach 1.08–4.30) [40] and hypersonic cone foredrag at MAPE 19.7% (eleven points, Mach 6.5–17.2, maximum +57%) [31]. The verification rests on documented, re-runnable benchmarks rather than on the author's own computational fluid dynamics; the comparators are published third-party datasets.

Third, we validated the integrated simulator against a reproducible 25-flight ground-truth corpus spanning Mach 0.54 to 4.33. The corpus inclusion rule is external: the flights are taken from Rogers' public RASAero II altitude-comparison set [4, 46] and comprise twenty-three single-stage flights and two two-stage flights (AeroPac 104K at Mach 3.04 and MESOS 293K at Mach 4.33), so the accuracy statistics are not outcome-curated by the present author. The mean signed apogee error is -0.38% (95% bootstrap confidence interval $[-2.41, +1.72]$, 20 000 resamples), with standard deviation 5.44%, RMSE 5.34%, MAE 4.74%, and all 25 flights within ±10% of measured apogee (14 within ±5%). The mean-error interval brackets zero, so the predictor is statistically unbiased on this corpus, and the bias-squared fraction of mean-square error is 0.01, indicating that the residual is essentially per-flight random scatter—build tolerance, motor-lot variation, and atmospheric soundings—rather than directional model bias. On the 25 paired flights the difference in absolute error against RASAero II is -0.60 percentage points (95% bootstrap CI $[-2.16, +0.96]$, straddling zero), and a Wilcoxon signed-rank test finds no significant difference ($W = 143.0$, $p = 0.615$) [51]. The honest claim is therefore *parity* with this version-locked RASAero II set, not superiority; the recorded RASAero II values are Rogers' archived predictions rather than independently rerun pre-flight cases. Because two base-drag scale constants are corpus-frozen, the headline is partly in-sample; a decontaminated prospective holdout that places every flight those constants touched into the development split gives a development MAE of 5.47% (n= 13) and a genuinely blind holdout MAE of 3.95% (n= 12). The holdout being *more* accurate than development is direct evidence that the two constants generalize rather than overfit.

B. Future Work

Concrete next steps, in priority order, follow from the disclosed limitations of Sec. X.

1) **Close the transonic blending bias.** The transonic regime (Mach 0.8–1.3) carries the largest per-regime signed error, -3.66%, indicating that component wave-drag contributions are presently underblended through the sonic line. Redistributing them through a Whitcomb equivalent-body-of-revolution area rule, and re-anchoring the blend against transonic range data, directly targets this bias.
2) **High-Mach validation with non-reconstructed vehicles.** The headline corpus reaches only Mach 4.33; the hypersonic regime is empty, and the exploratory high-Mach sounding-rocket results (Sec. IX) are dominated by motor- and geometry-reconstruction uncertainty on poorly documented historical flights. Admitting sounding rockets on merit requires ground-truth vehicles with independently documented thrust curves and geometry, not reconstructions, so that an above-Mach-5 integrated claim rests on measurement rather than on inferred inputs.
3) **Recover the MESOS regression.** The two-stage MESOS 293K closure predicts -6.96% (273,056 ft) from the current archived code release—a disclosed regression relative to the earlier RFD v1.2 snapshot (291,601 ft, -0.6%), which was produced by an earlier code version. The -6.96% value is itself reproducible from the present release, and the flight remains within the ±10% band; the root cause is under investigation, and the next RFD release will re-sync the model column. Isolating and recovering this regression is required before the two-stage staging-and-coast model can be presented as fully validated.

4) **Anchor the pitch-damping multiplier to wind-tunnel data.** The transonic pitch-damping (C_{m_q}) augmentation relies on a single 3× multiplier that overshoots reference computations by 110–160% per Sznajder [37] and is held at B-level. Lowering it to a wind-tunnel-anchored value, validated at flight scale, would raise this subsystem to A-level confidence.
5) **Ship closed-loop CFD comparators.** The CFD validation (Sec. VII) cites four published third-party datasets rather than runs the author can drive end to end. Building an own AGARD-B model file to drive the Vidanović SST k–ω reference [45] and an own RM-10-class ogive–cylinder–boattail file to drive the Sahu base-drag reference [44] would remove the "reference dataset only" caveat.

All source code, the 25-flight Rocket Flight Database [10], and the analysis scripts that regenerate every figure and table are released under permissive licenses for end-to-end reproduction. Community contributions on any of the above items—particularly the transonic blending fix and the MESOS regression—are warmly invited through the open-source release.

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Data and Code Availability

The Rocket Flight Database that defines the ground-truth corpus—measured apogees, vehicle and motor metadata, and the per-flight RASAero II reference predictions—is released under the Creative Commons Attribution 4.0 International (CC-BY-4.0) license and archived at Zenodo, DOI 10.5281/zenodo.19976138 [10], with an actively maintained mirror at `https://github.com/AidanSYu/rocket-flight-database`. The OpenRocket-Plus source code that implements the shock-geometry pre-pass and the downstream supersonic models is available at `https://://github.com/AidanSYu/openrocketsupersonic`; a tagged release corresponding to the results reported here, together with an archival code Zenodo deposit (DOI `10.5281/zenodo.XXXXXXX`), will be minted at submission. The per-flight OpenRocket-Plus predictions reported here are regenerated from that tagged code release; for the MESOS 293K closure the present release predicts −6.96%, a disclosed regression from the −0.6% recorded in the archived database v1.2 snapshot (reproducible in isolation and under investigation, §X). A synchronized database release carrying this model column will be deposited alongside the code archive at submission, so that the headline statistics regenerate exactly from the cited record. The analysis scripts `analyze.py` and `uncertainty_quantification.py`, distributed with the database repository, regenerate every figure, summary statistic, and bootstrap confidence interval reported in this paper directly from the published comparison CSV, enabling end-to-end reproduction of the validation results.

AI Disclosure

In accordance with the AIAA editorial policy on the use of artificial intelligence (October 2024), the author discloses that a generative AI assistant (Anthropic Claude) was used for language editing, manuscript formatting, and code review of the analysis scripts. All text, scientific claims, equations, and numerical results were authored, derived, and verified by the human author, who takes full responsibility for the content. No AI system is listed as an author or qualifies for authorship.

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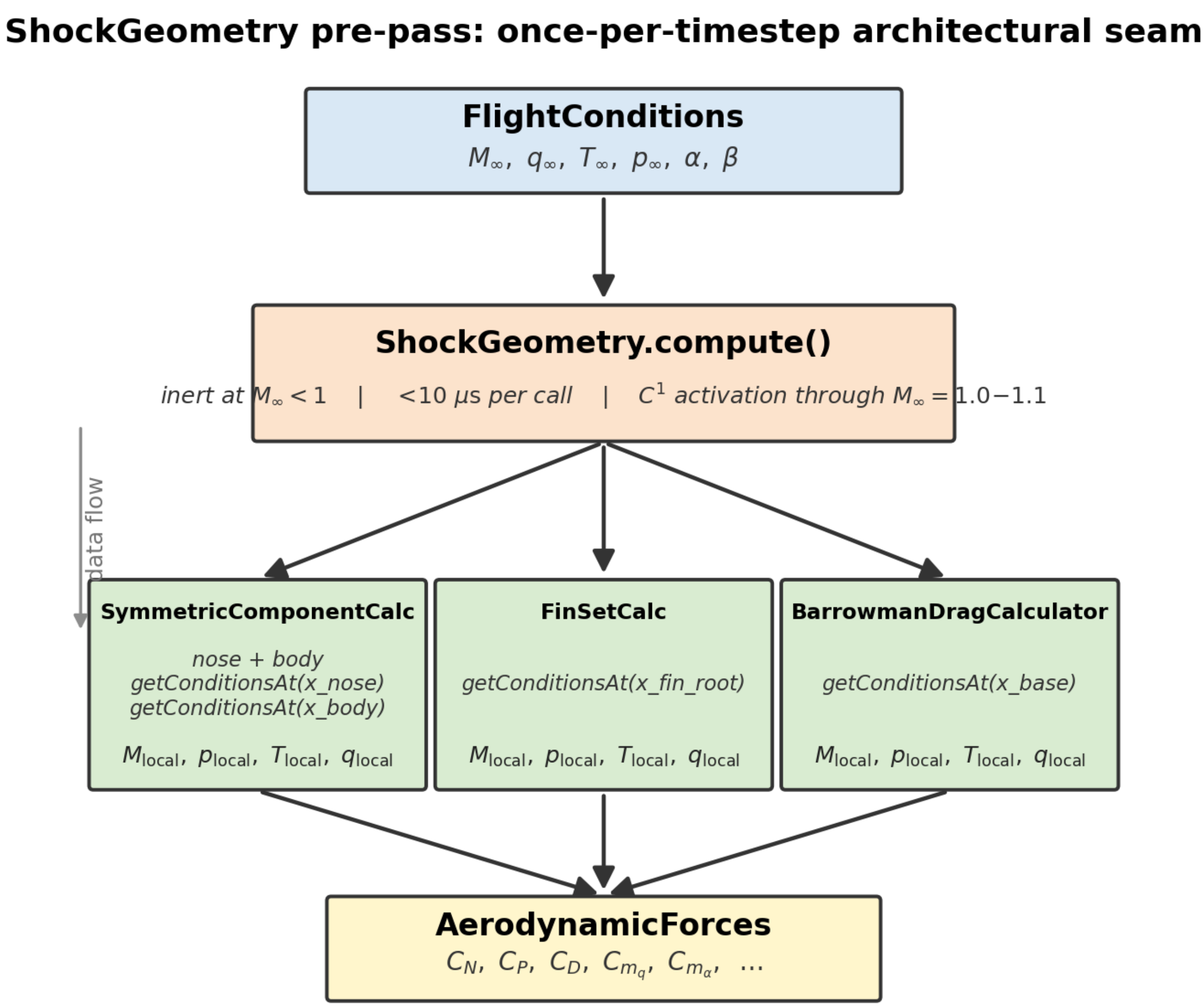

Fig. 2 Shock-geometry pre-pass data-flow block diagram. Once per timestep the pre-pass surface-marches the airframe and publishes the local post-shock Mach, pressure, and temperature; each component calculator queries the local state at its axial station before computing forces and moments.

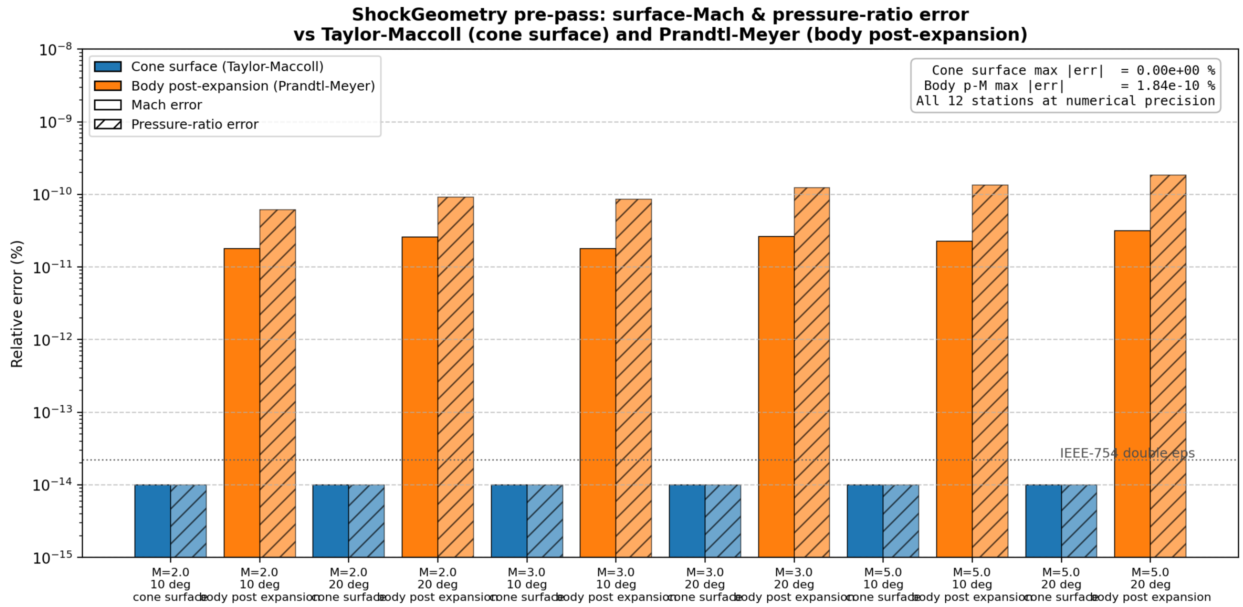

Fig. 3 Shock-geometry surface-Mach verification against Taylor–Maccoll cone flow and Prandtl–Meyer expansion at six (M_∞, θ_c) reference points spanning $M_\infty = 1.5$–5.0 and $\theta_c = 10°$–$20°$. Cone-surface Mach is reproduced to 0.00% and shoulder-expansion Mach to better than $4 \times 10^{-11}\%$.

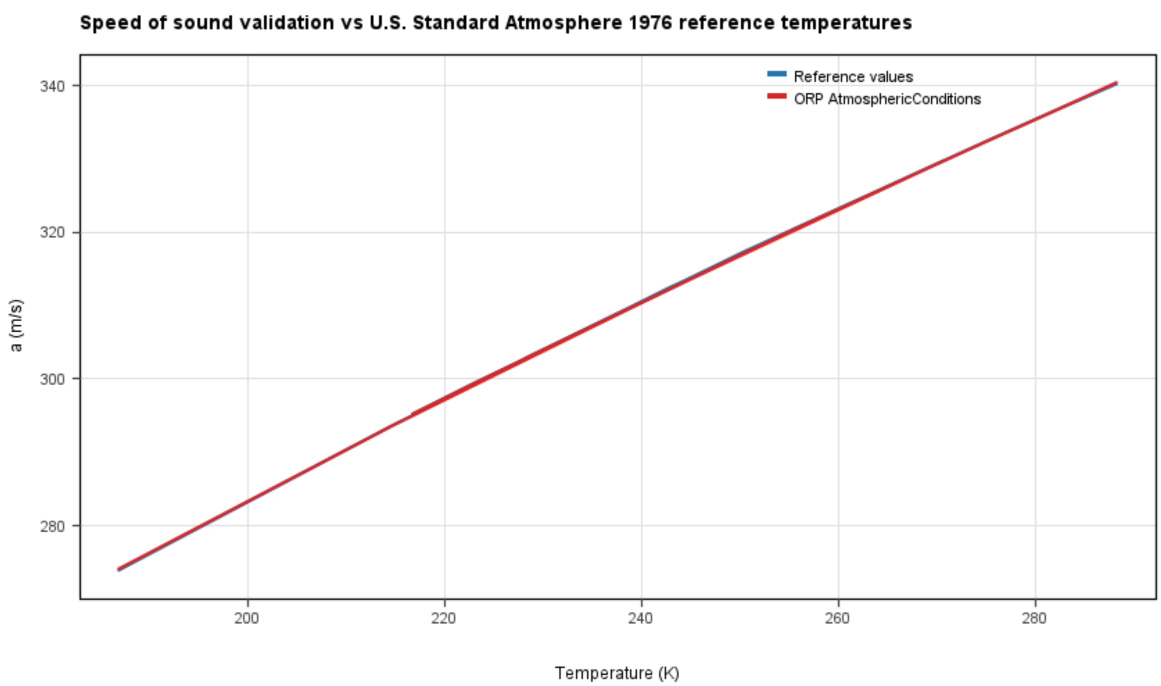

Fig. 4 Speed of sound from the analytical relation $a = \sqrt{\gamma R T}$ versus the tabulated U.S. Standard Atmosphere 1976 profile over twenty altitudes from sea level to $80\,\mathrm{km}$; maximum error $0.016\,\%$.

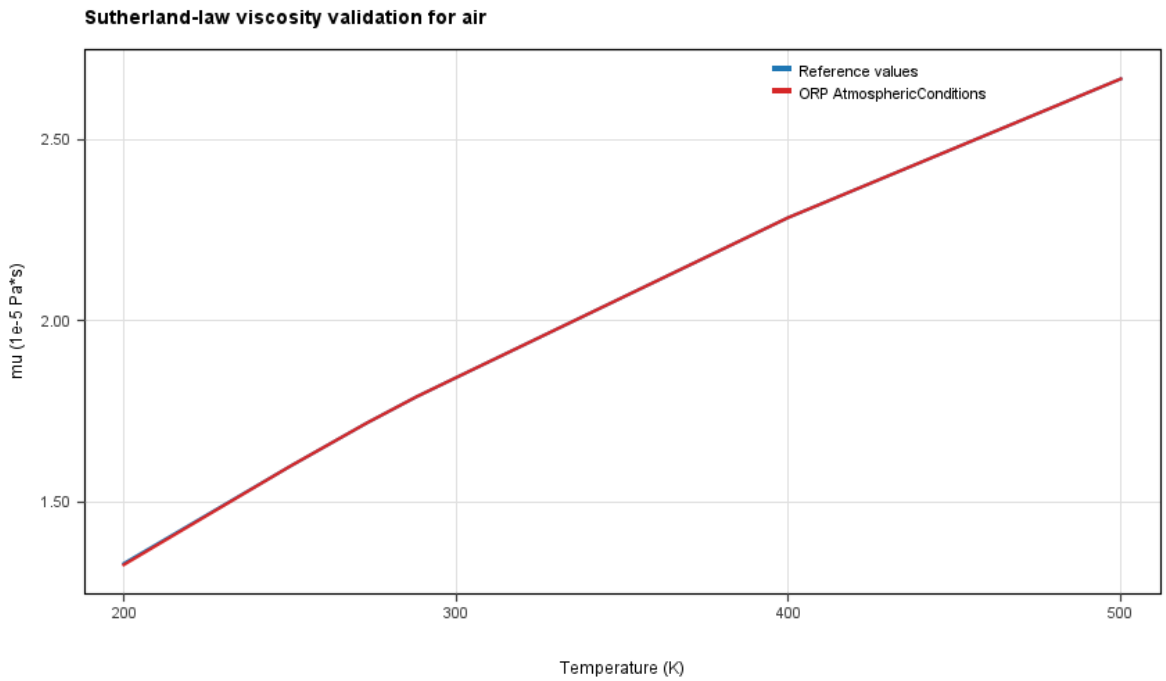

Fig. 5 Sutherland viscosity, Eq. (3), versus reference air-property tables over 150 K to 500 K; MAPE 0.54 % with no systematic bias.

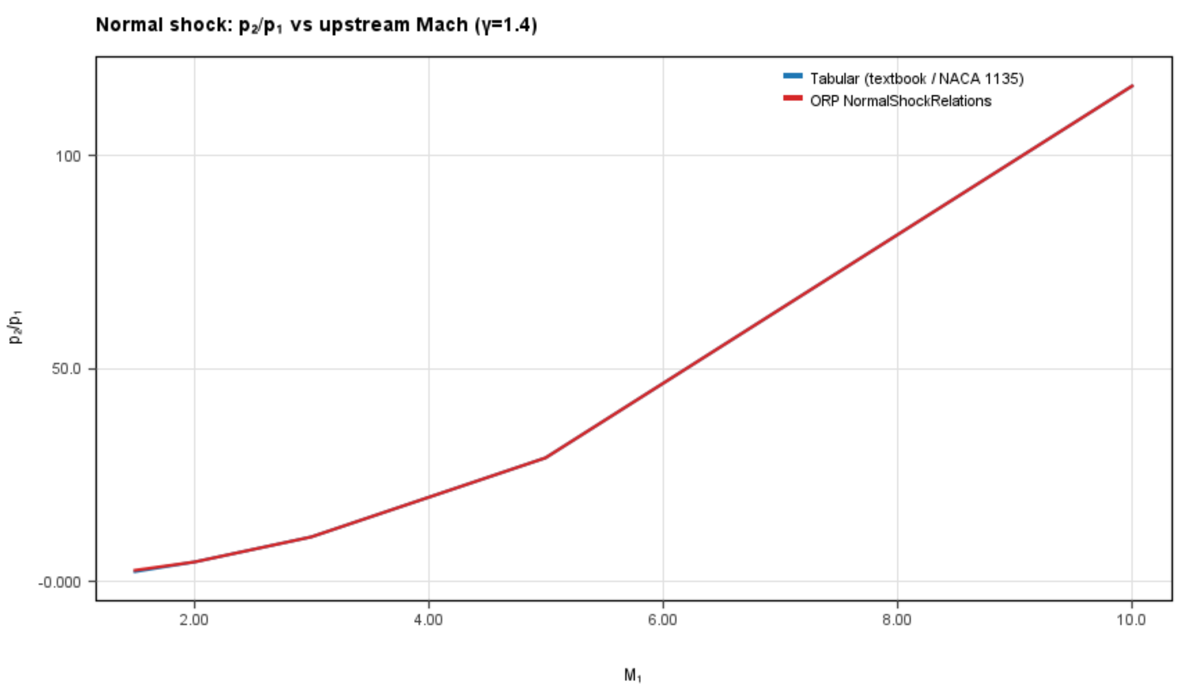

Fig. 6 Normal-shock pressure ratio p_2/p_1, Eq. (7), versus NACA Report 1135 over $M = 1.5$–10.

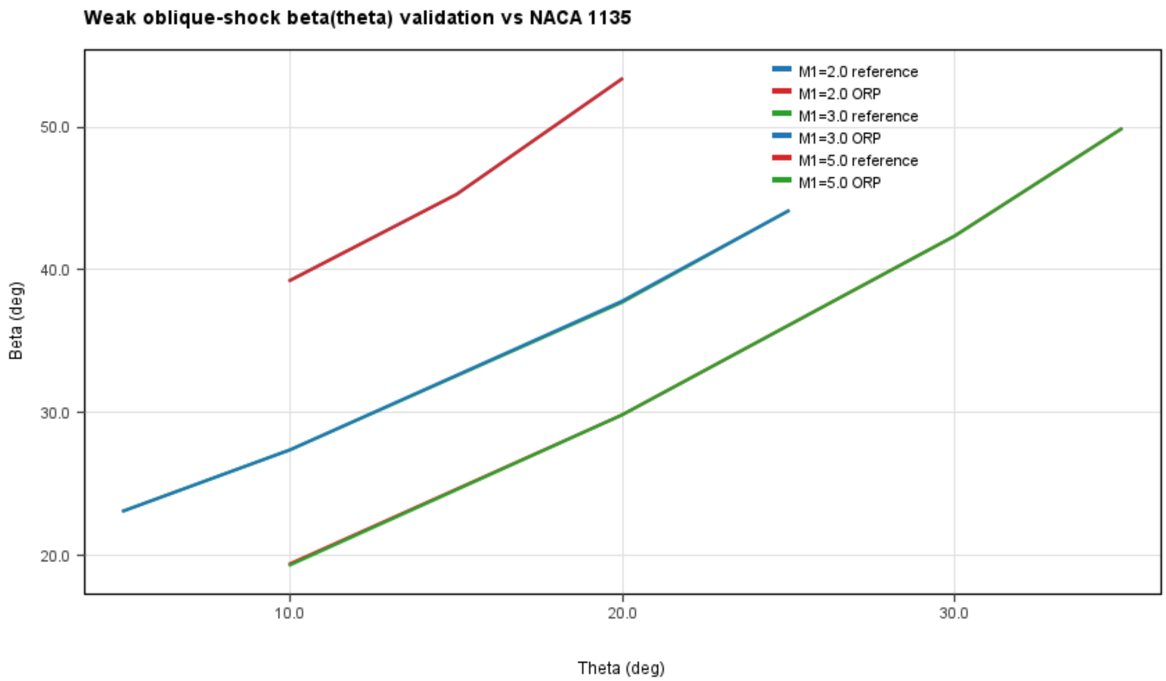

Fig. 7 Oblique-shock angle $\beta(\theta)$ from the θ–β–M relation, Eq. (9), versus NACA Report 1135 at $M = 2.0$, 3.0, and 5.0.

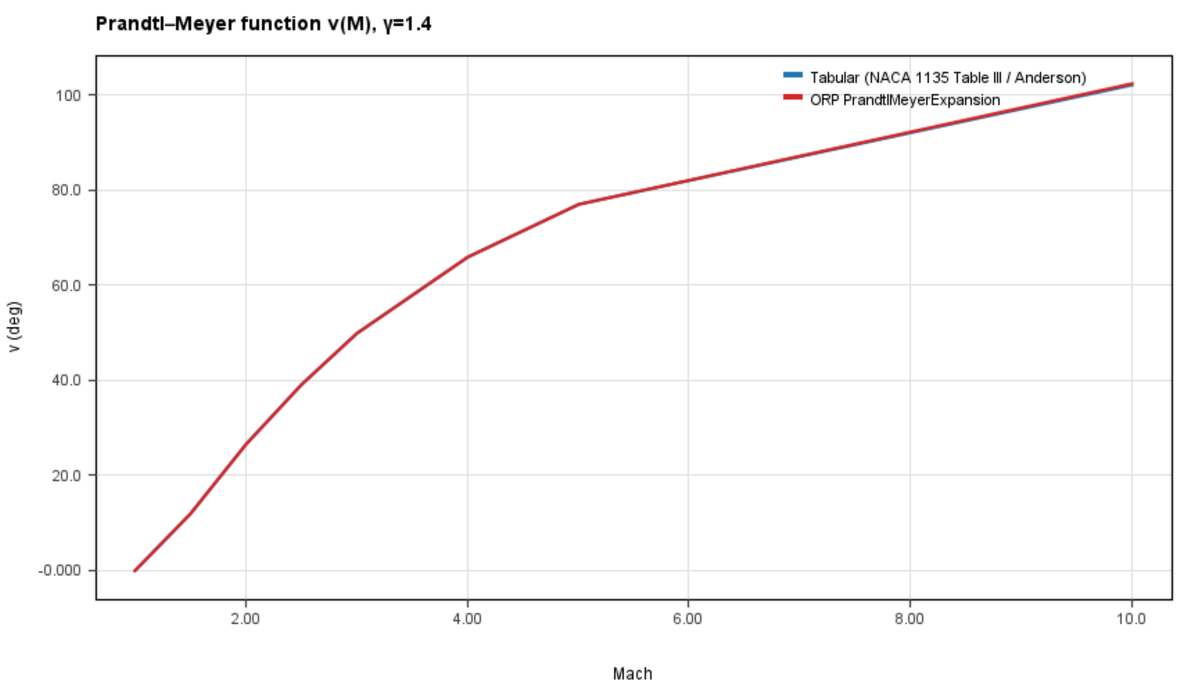

Fig. 8 Prandtl–Meyer function $\nu(M)$, Eq. (11), versus NACA Report 1135 Table III.

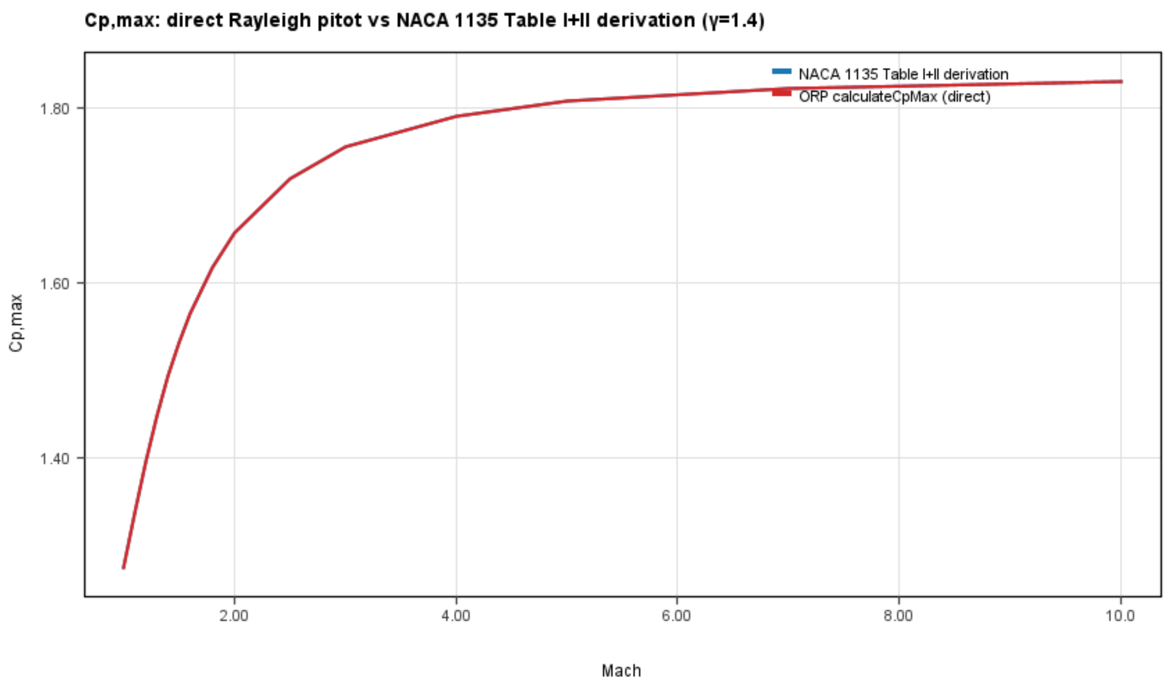

Fig. 9 Rayleigh pitot $C_{p,\max}$, Eq. (8), versus NACA Report 1135 at fifteen Mach points, $M = 1$–10.

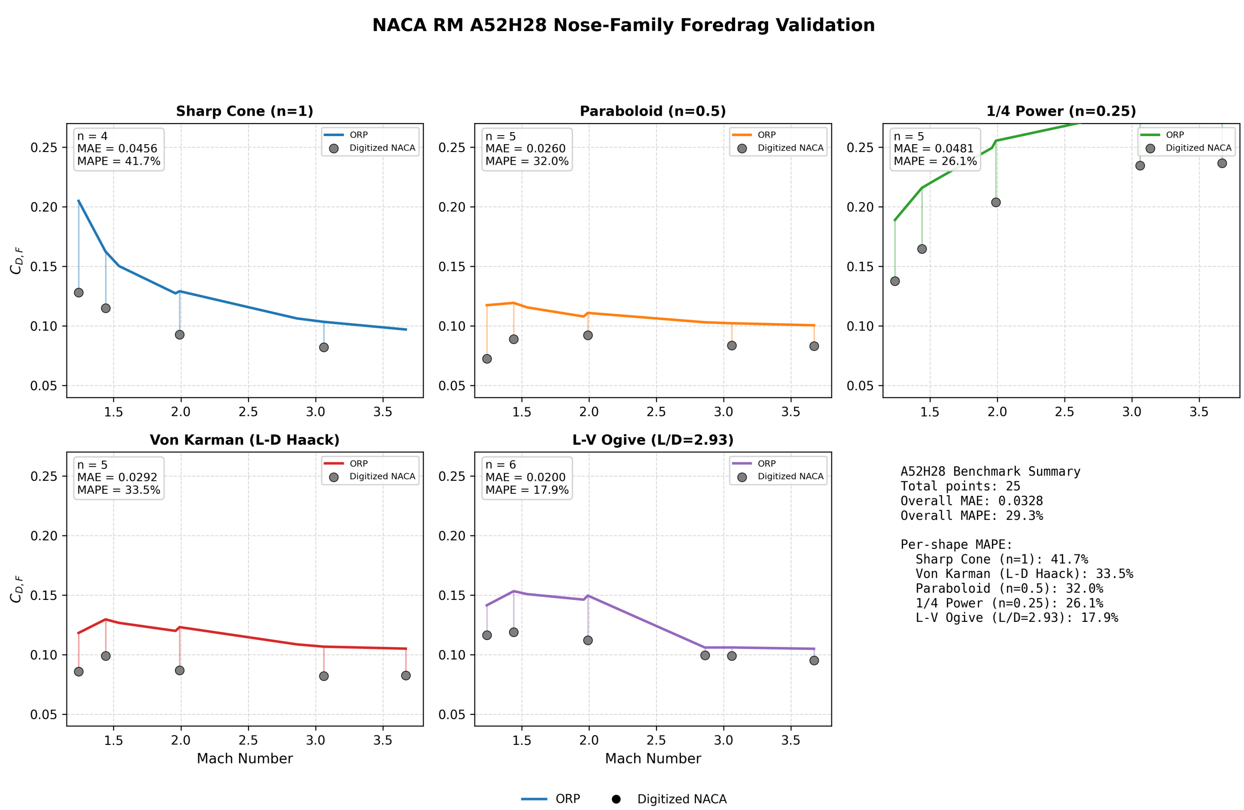

Fig. 10 Nose wave drag versus NACA RM A52H28 [22] for cone, quarter-power, three-quarter-power, Haack, and L-V ogive noses at fineness 3, $M = 1.5$–3.0. Aggregate mean absolute error = 0.029 in C_d.

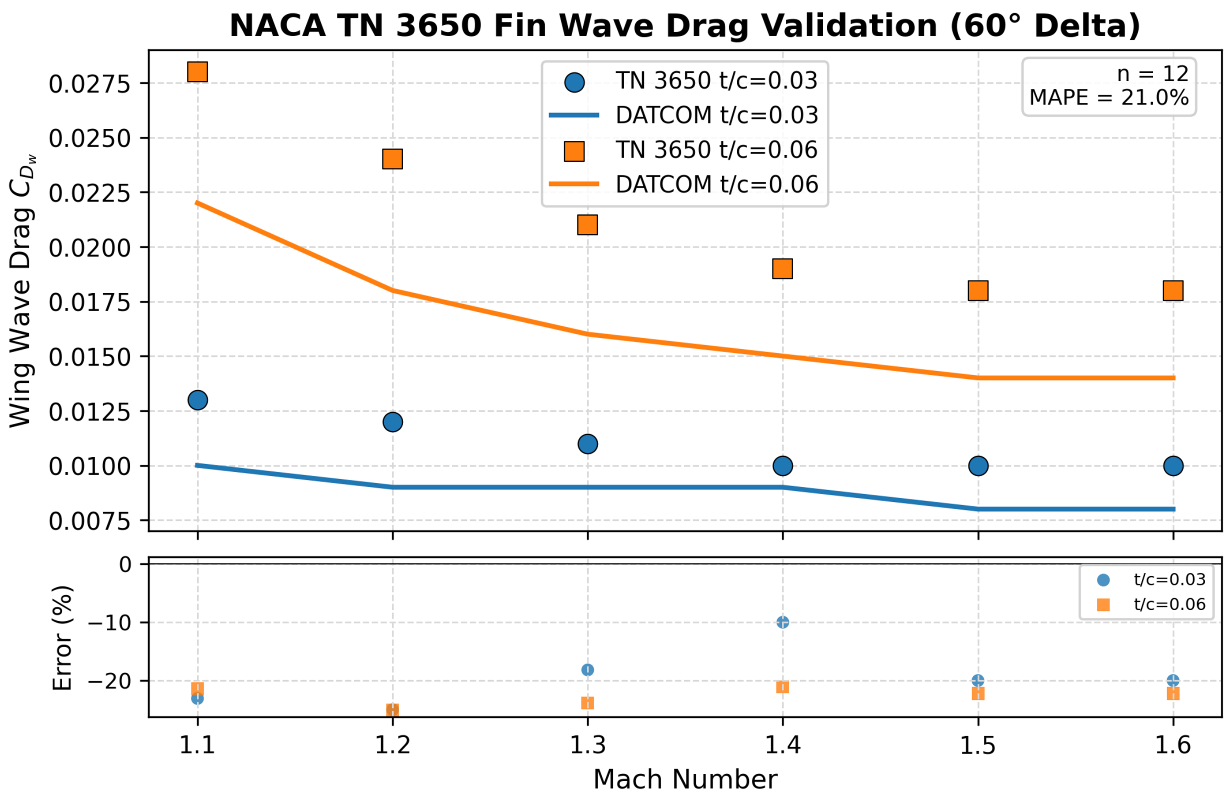

Fig. 11 Fin wave drag for a 60-deg delta wing: present DATCOM 4.1.5.1 implementation versus NACA TN 3650 [24] free-flight data. Mean absolute percentage error = 21.0% over 12 points, $M = 1.1$–2.5.

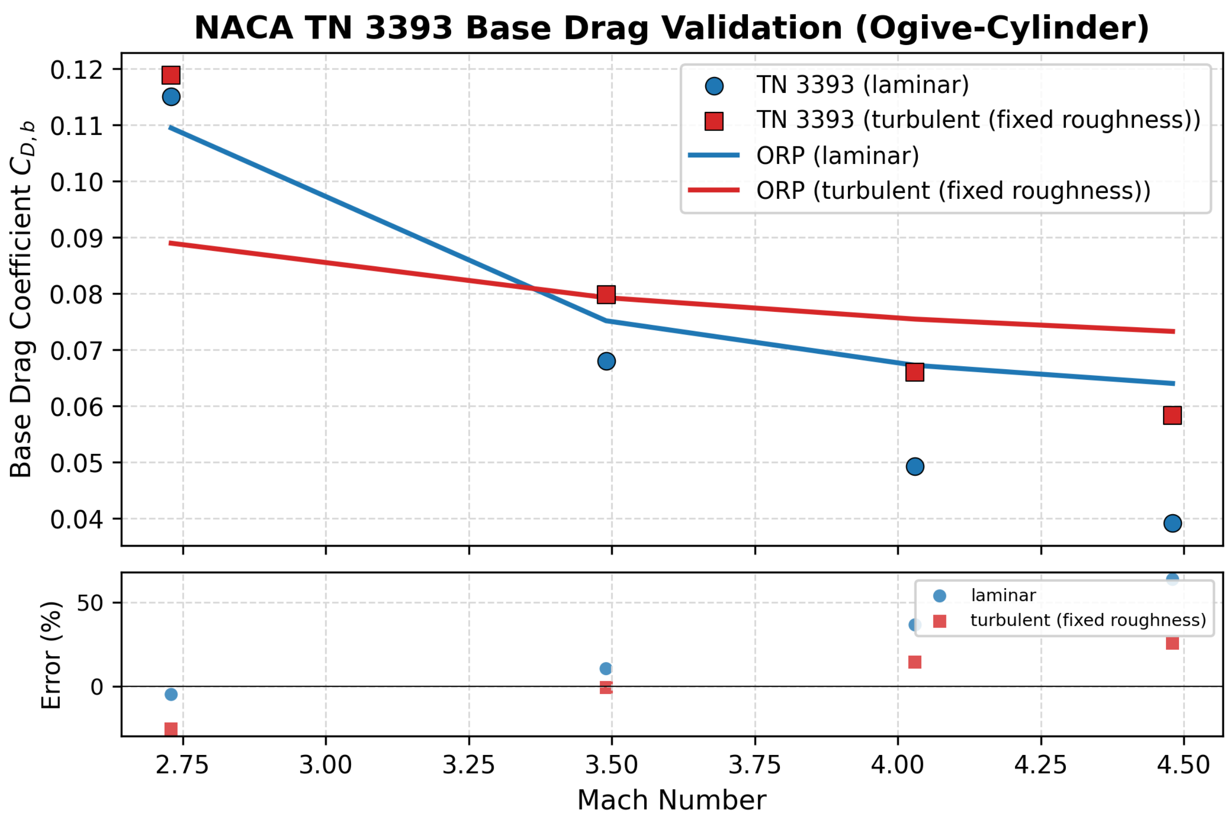

Fig. 12 Turbulent and laminar base drag versus NACA TN 3393 [26]. The supersonic turbulent correlation, Eq. (19)**, attains a mean absolute percentage error of** 15.9%**; the Chapman laminar correlation [27], Eq.** (20)**, attains** 4.4%**.**

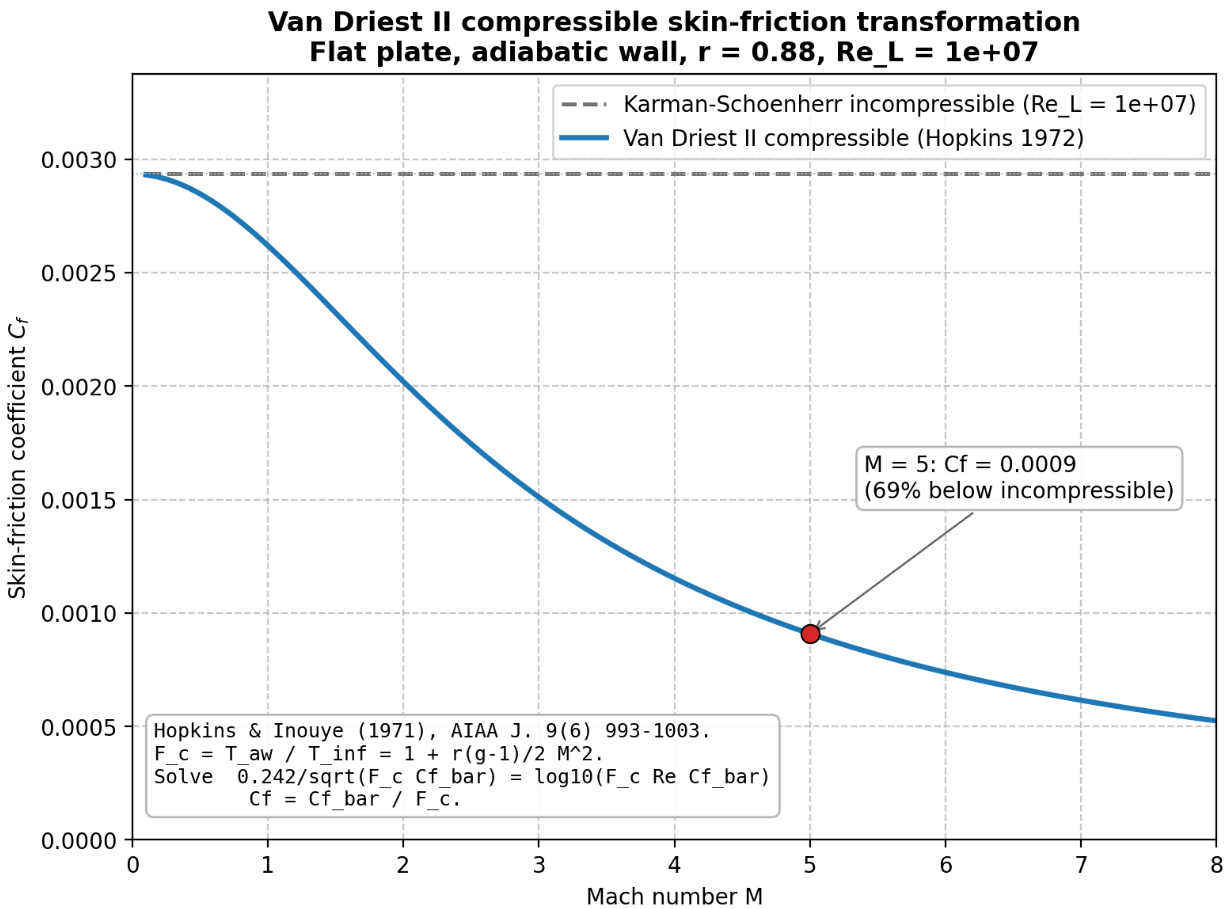

Fig. 13 Van Driest II compressible skin-friction coefficient versus Mach number for a representative slender body at length Reynolds number $\mathrm{Re}_L = 10^7$, after the Hopkins–Inouye formulation [18].

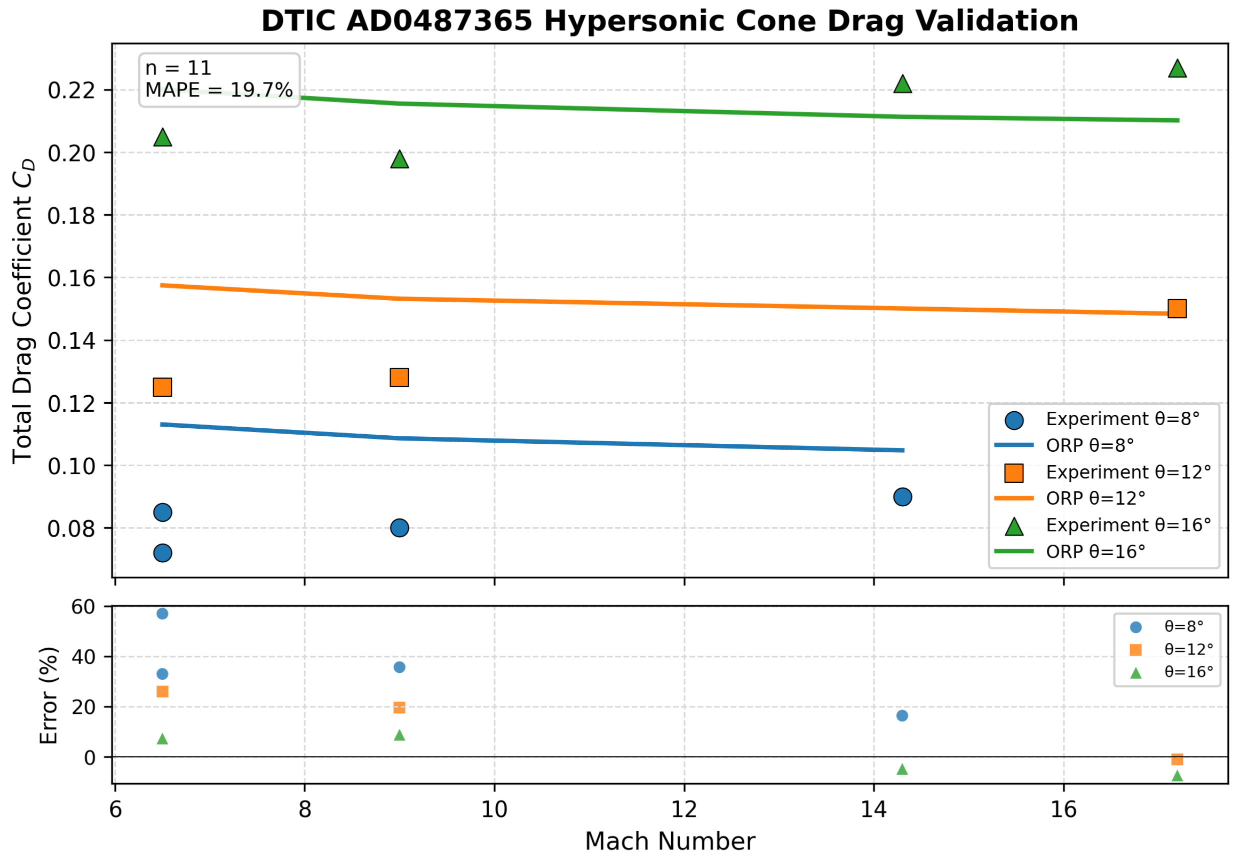

Fig. 14 **Hypersonic cone foredrag versus DTIC AD0487365 [31] ballistic-range data, $M = 6.5$–17.2, half-angles 8, 12, and 16 deg. Aggregate mean absolute percentage error = 19.7% over 11 points; the largest residual is $+57\%$ for the thinnest cones at the lowest Reynolds number.**

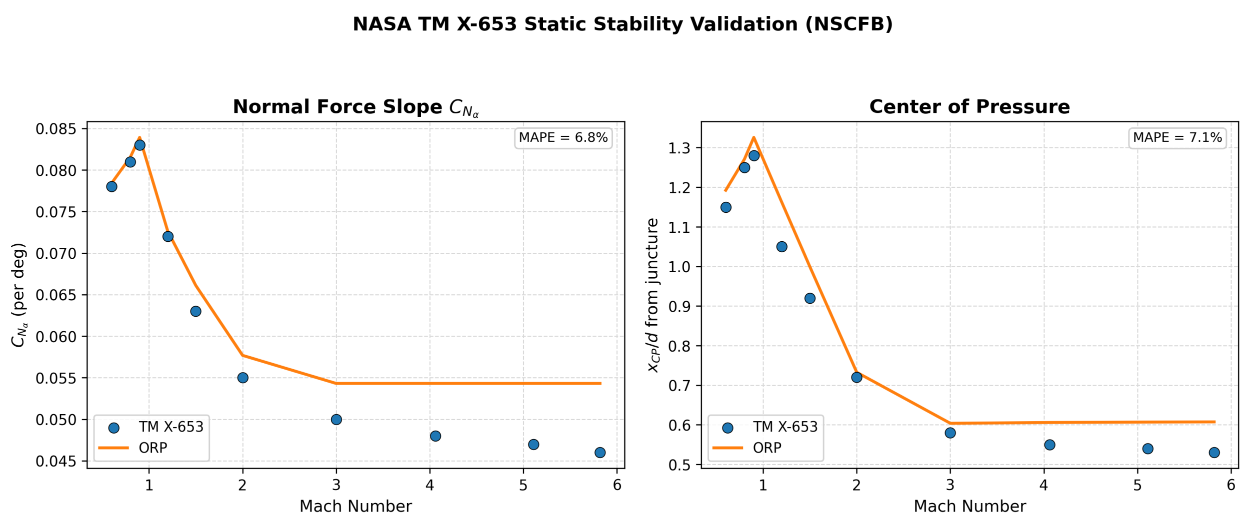

Fig. 15 **Static stability against the NASA TM X-653 wind-tunnel database [34], $M = 0.6$–5.82: finned-body normal-force slope C_{N_α} and center-of-pressure location x_{CP}/d. The local-flow model achieves C_{N_α} MAPE 6.84% and x_{CP}/d MAPE 7.11% over ten test points each, with the largest residual confined to the transonic band.**

Table 4 Drag submodel inventory (present work).

Submodel	Regime	Source / formulation	Validation source	Metric
Taylor–Maccoll cone pressure	$M = 1$–17	Exact conical-flow ODE (Sec. III)	NACA RM A52H28 cone case	Exact (analytic)
Shock-expansion ogive	$M = 1.3$–10	Eq. (16), 100-strip integrator	NACA RM A52H28 [22]	MAE 0.029 (5 shapes)
Dahlem–Buck shape factor	$M = 1.3$–6	Eq. (17)	NACA RM A52H28 [22]	K_{shape} within 5%
Fin wave drag, DATCOM 4.1.5.1	$M = 0.9$–$5+$	Eq. (18)	NACA TN 3650 [24] / Ackeret	MAPE 21%; analytic 0.00%
Supersonic turbulent base	$M > 1.3$	Eq. (19)	NACA TN 3393 [26] (turb.)	MAPE 15.9%
Chapman laminar base	$M = 1.3$–4.5	Eq. (20)	NACA TN 3393 [26] (lam.)	MAPE 4.4%
Chapman–Korst shear layer	$M = 1.2$–1.4	ESDU 77021 [25] BL correction	Blend region	Continuity
Viswanath boattail	Any M, boattail	$\Delta C_{d,b}(\theta_{bt}, M)$	Viswanath [28]	15–40% reduction
Power-on base	During burn	$k_{po}(A_e/A_{\text{ref}}, p_e/p_\infty)$	NASA SP-8039 [29]	Qualitative
Van Driest II C_f	$M = 1.1$–9	Eqs. (12)–(14) (Sec. III), (21)	Hopkins & Inouye [18]	~50% C_f at $M = 5$
Modified Newtonian	$M > 5$	$C_p = C_{p,\max} \sin^2\theta$	DTIC AD0487365 [31]	MAPE 19.7%

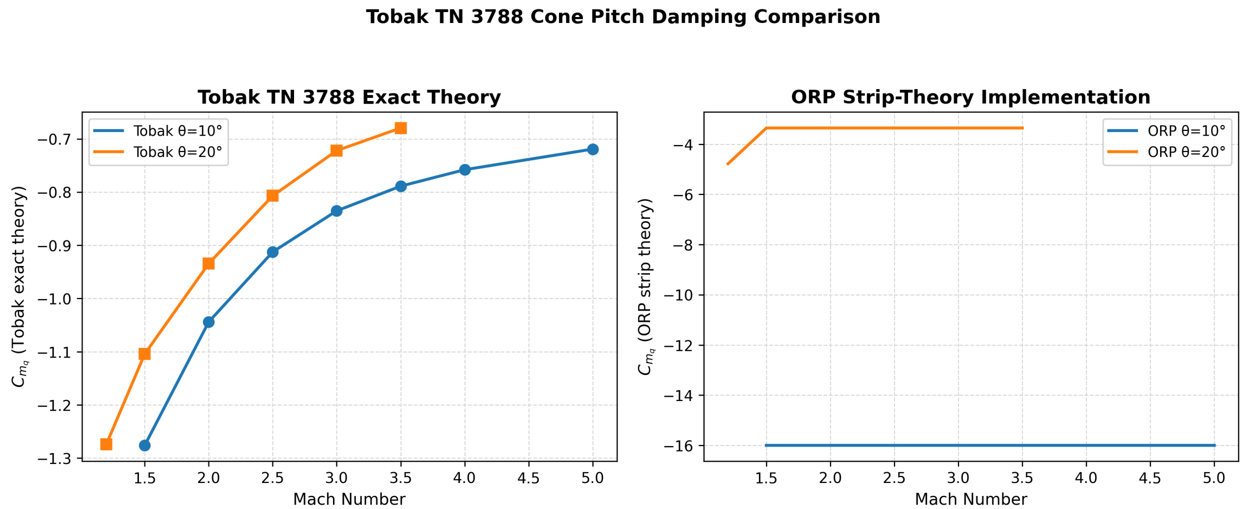

Fig. 16 Strip-theory pitch-damping derivative C_{m_q} compared with the exact slender-body theory of Tobak and Wehrend, NACA TN 3788 [36]. The sign is correct and the damping is conservative, but strip theory overpredicts the magnitude for an isolated axisymmetric body by a factor of five to ten; the derivative is therefore reported as a conservative engineering estimate rather than a quantitative result.

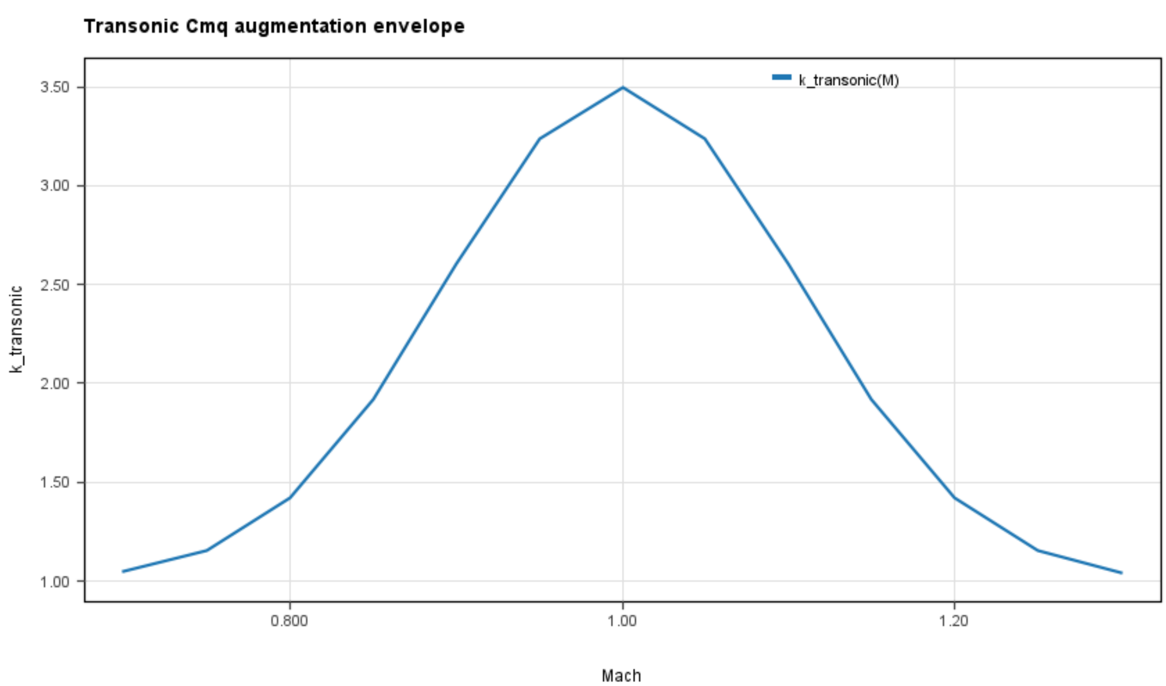

Fig. 17 Transonic Gaussian augmentation $k_{\text{transonic}}(M)$ of the pitch-damping derivative, Eq. (27), with peak augmentation parameter $k(1.0) = 3.5$ at $M = 1$ and decaying to unity within ± 0.3 Mach. The accompanying $3\times\ C_{m_q}$ multiplier is a B-level adjustment calibrated against the AEDC-TR-76-58 transonic pitch-damping wind-tunnel data with no independent flight-scale anchor; at $M = 1.08$–1.11 the augmented model overshoots the physical transonic damping peak resolved by the Sznajder CFD by an estimated 110–160% [37].

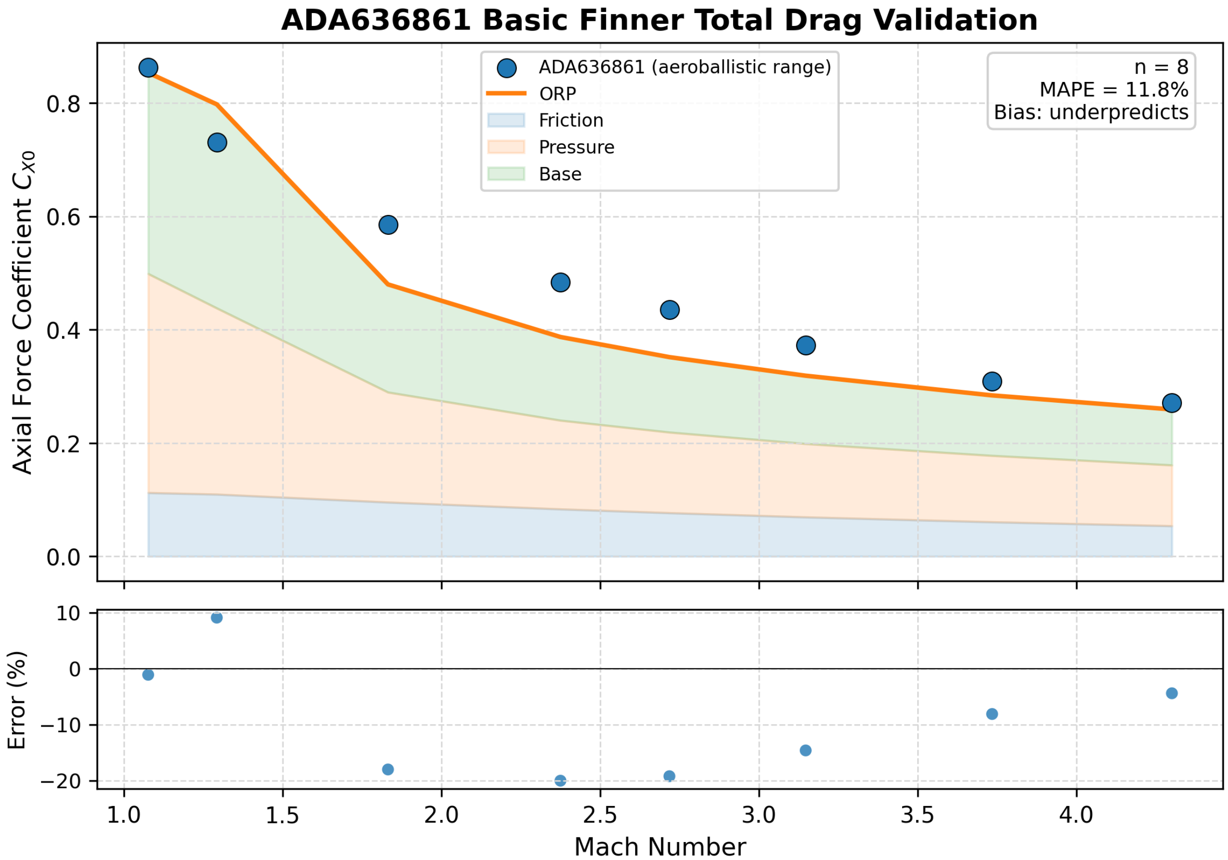

Fig. 18 Finned-vehicle total drag for the Army–Navy Basic Finner reference projectile versus the Dupuis and Hathaway ballistic-range data [40], M = 1.08–4.30**. Aggregate MAPE** 11.8% **over eight points.**

Table 6 Published-CFD comparator inventory.

Source	Geometry	Quantity	Mach range	Comparison status
Bunescu et al. [43] (URANS)	Basic Finner	C_X	0.4–3.5	Comparator wired; trend correct, loose absolute agreement
Sahu et al. [44] (thin-layer NS)	Ogive-cylinder-boattail	$C_{D,b}$, C_D	0.9–1.2	Memo only; digitization deferred to future work
Vidanović et al. [45] (SST k–ω)	AGARD Model B	C_D, C_L, C_m	0.596, 1.602	Reference dataset; no AGARD-B build shipped
Sznajder [37] (Fluent)	Basic Finner	$C_{m_q} + C_{m_{\dot{\alpha}}}$	0.9–5.0	Comparator wired; supersonic MAPE 31.6% (n = 8, $M \geq 1.29$)
Bhagwandin & Sahu [42] (CFD++)	AFF / ANF finners	$C_{m_q} + C_{m_{\dot{\alpha}}}$	1.3–4.5	Second source; supersonic MAPE 19.0% (AFF, n = 5), 28.0% (ANF, n = 8)

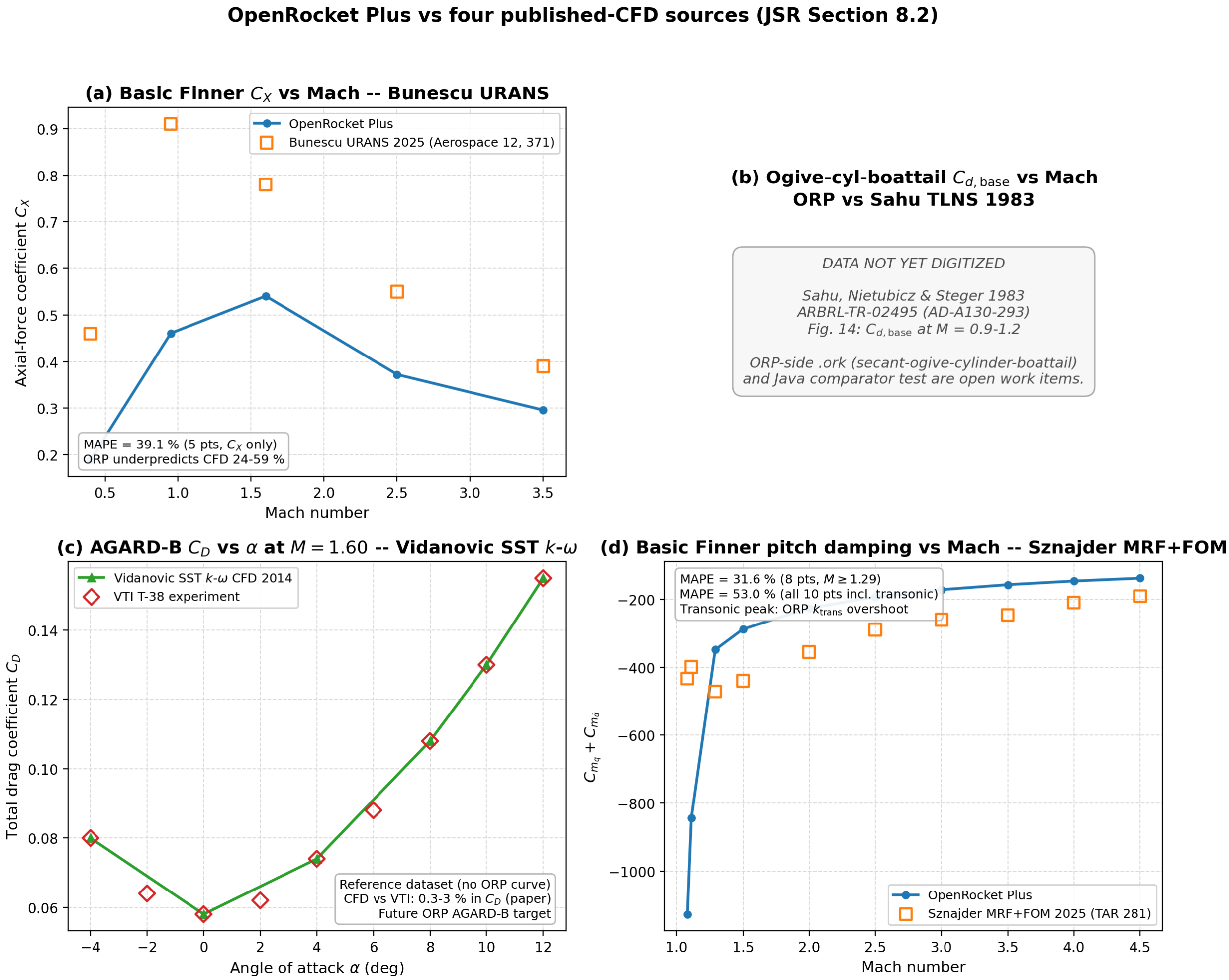

Fig. 19 Published-CFD comparator composite. Panel A: Basic Finner axial-force coefficient versus the Bunescu URANS solution [43] (correct trend, loose absolute agreement). Panel B: ogive-cylinder-boattail base drag versus the Sahu thin-layer Navier–Stokes solution [44] (digitization deferred). Panel C: AGARD Model B reference dataset versus the Vidanović SST k–ω solution and VTI T-38 experiment [45]. Panel D: Basic Finner pitch-damping sum versus the Sznajder Fluent solution [37], cross-corroborated on the AFF and ANF finner planforms by the CFD++ study of Bhagwandin and Sahu [42].

Table 7 Headline accuracy on the externally defined 25-flight corpus (Mach 0.54 to 4.33). RASAero II values are Rogers' recorded predictions for the same flights.

Predictor	N	Mean	σ	RMSE	MAE	$\leq \pm 5\%$	$\leq \pm 10\%$
OpenRocket-Plus	25	−0.38	5.44	5.34	4.74	14/25	25/25
RASAero II	25	+2.46	5.81	6.20	5.34	13/25	22/25

All entries in percent except N and the count columns.

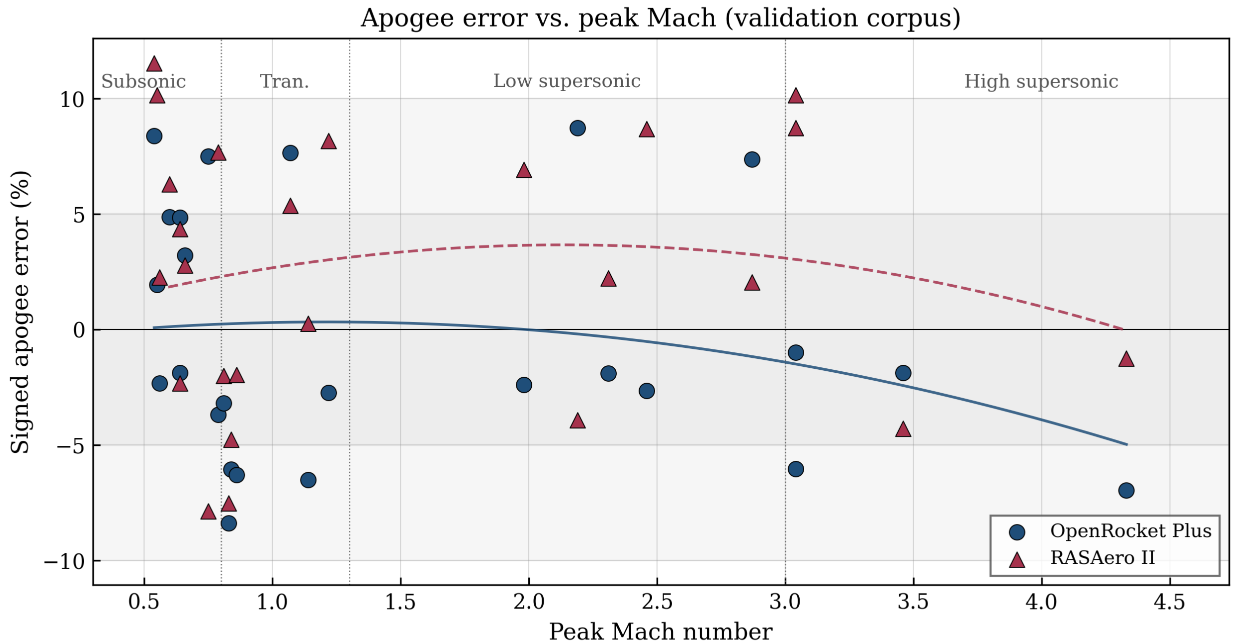

Fig. 20 Apogee-prediction error versus peak Mach for the 25-flight corpus. Residuals remain bounded within $\pm 10\,\%$ across the full subsonic-to-supersonic range; the transonic band carries the largest negative bias.

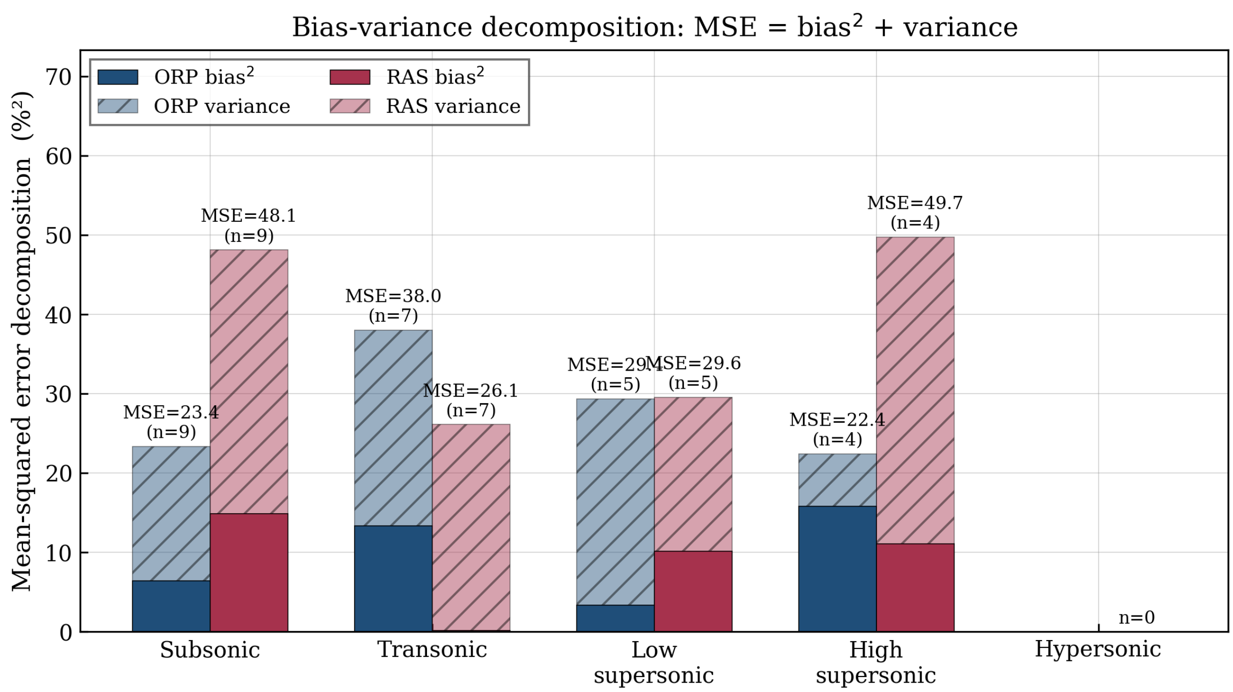

Fig. 21 Bias–variance decomposition of the mean squared apogee error. For OpenRocket-Plus the squared-bias term is a negligible fraction ($\mathrm{bias}^2/\mathrm{MSE} = 0.01$) of the total, indicating an essentially unbiased predictor whose residual is dominated by variance.

Table 8 Per-regime ORP accuracy on the 25-flight corpus. Biases, σ, RMSE, and MAE in percent; count columns are flights within tolerance.

Regime	Mach	N	Bias	σ	RMSE	$\leq 5\%$	$\leq 10\%$
Subsonic	< 0.8	9	+2.54	4.37	4.83	7	9
Transonic	0.8–1.3	7	−3.66	5.36	6.17	2	7
Low supersonic	1.3–3.0	5	+1.83	5.70	5.42	3	5
High supersonic	3.0–5.0	4	−3.98	2.97	4.73	2	4
Hypersonic	> 5.0	0	—	—	—	—	—

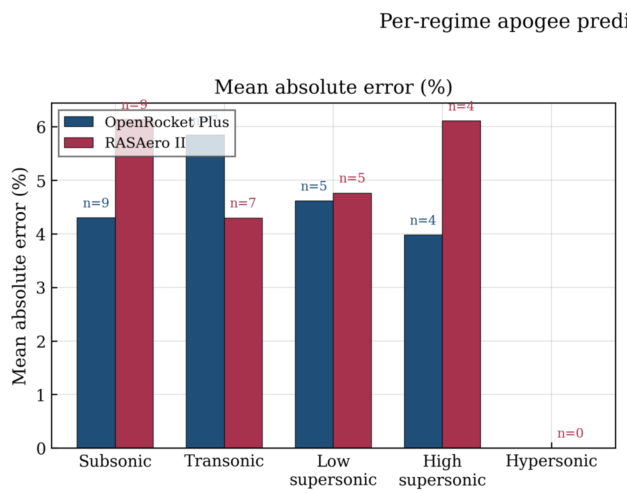

Fig. 22 Per-regime error distribution. The transonic band carries the largest negative bias; the high-supersonic band has the tightest scatter. Bands above Mach 3 contain four or fewer flights and are descriptive only.

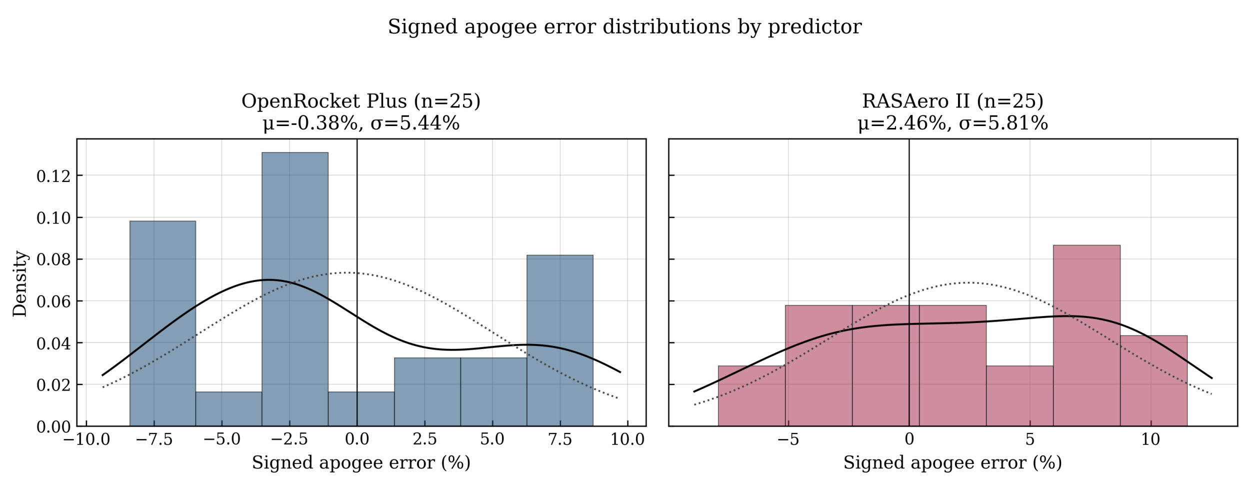

Fig. 23 Error distributions for OpenRocket-Plus and RASAero II on the 25-flight corpus. Both are light-tailed; the ORP distribution centers near zero.

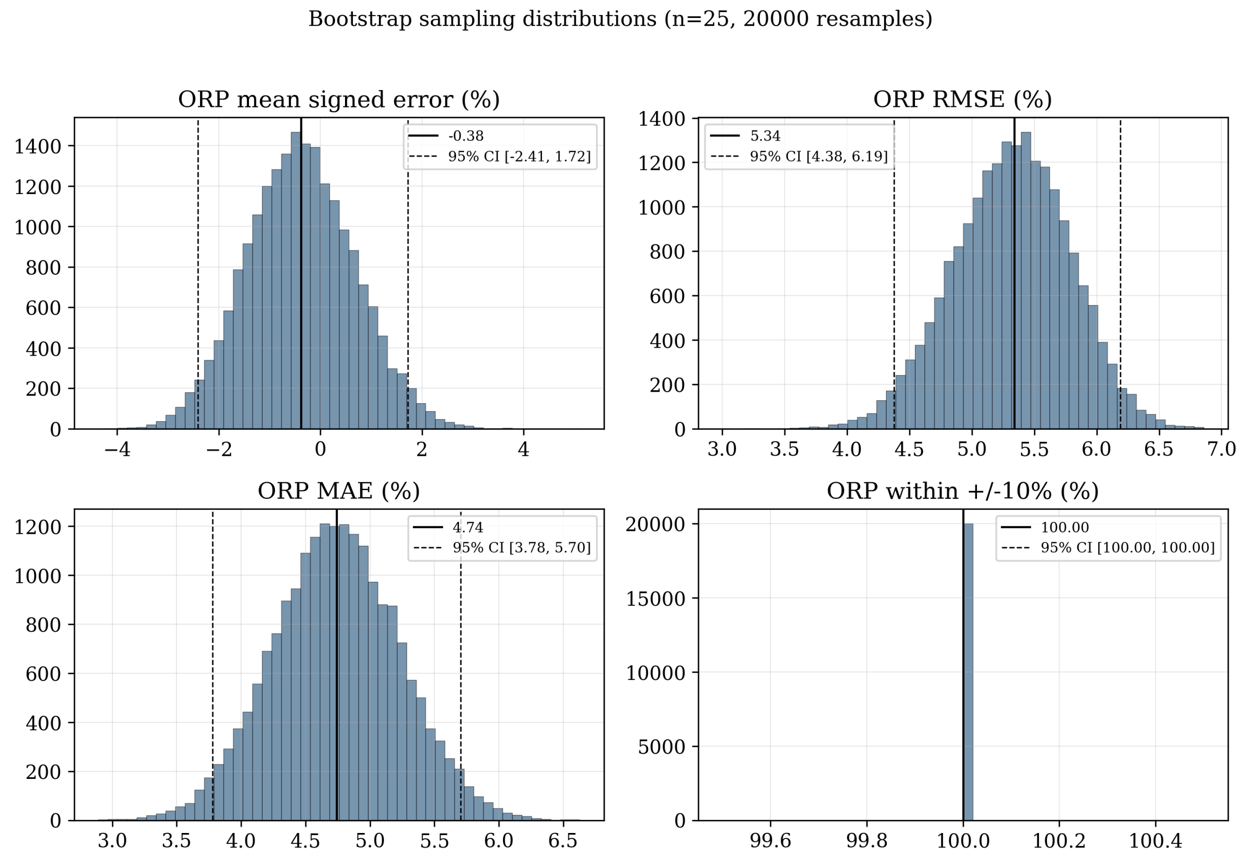

Fig. 24 Bootstrap sampling distributions (20 000 **resamples, seed `0x51A7EA`) for the headline ORP statistics. The mean-error distribution straddles zero; no bias is detectable at this sample size.**

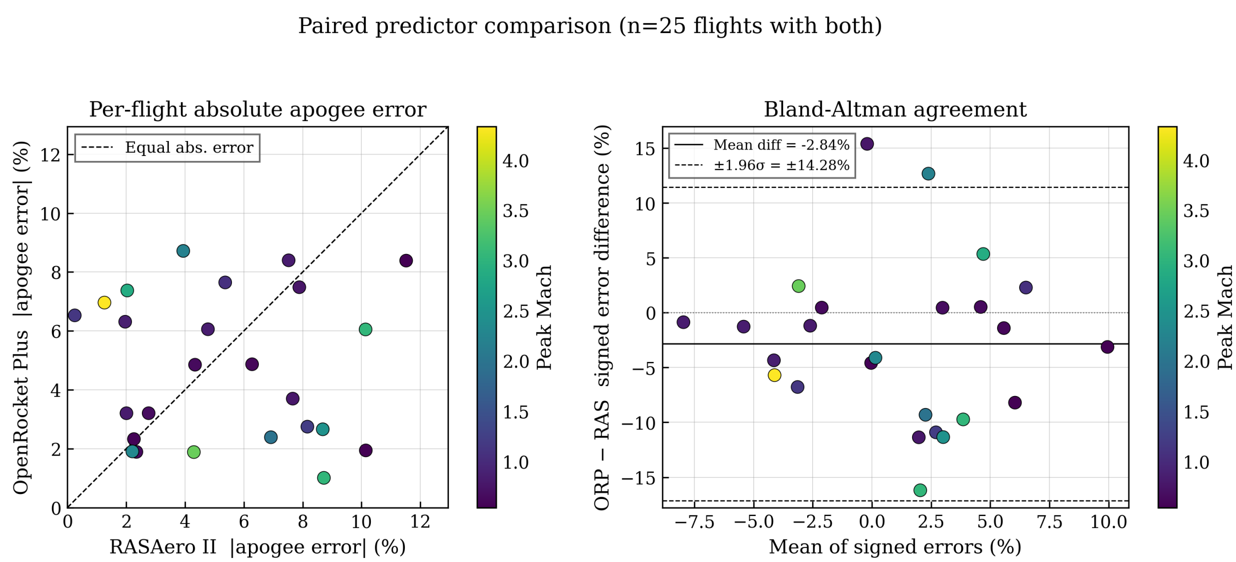

Fig. 25 Paired per-flight comparison of OpenRocket-Plus and RASAero II apogee errors. The paired absolute-error difference is statistically indistinguishable from zero (Wilcoxon $p = 0.615$**), supporting a parity conclusion.**

Table 9 Per-flight results for the 25-flight corpus, ordered by peak Mach. Apogees in feet; errors in percent of measured apogee. RASAero II values are Rogers' recorded predictions for the same flights.

	Vehicle	Motor	M_{pk}	h_{meas}	h_{RAS}	h_{ORP}	ε_{RAS}	ε_{ORP}
1	Thunder & Lightning	I284W	0.54	3577	3989	3877	+11.52	+8.4
2	Gibb	I284W	0.55	3913	4310	3989	+10.15	+1.9
3	Cancer Descending	M1297W	0.56	6188	6328	6044	+2.26	−2.3
4	EZI-65 J450ST	J450ST	0.60	3965	4214	4158	+6.28	+4.9
5	Caliber Isp 04 AVTC Team 2	I205	0.64	3710	3871	3890	+4.34	+4.9
6	Caliber Isp 04 AVTC Team 3	I205	0.64	3964	3871	3889	−2.35	−1.9
7	Caliber Isp 04 AVTC Team 1	I205	0.66	3837	3943	3960	+2.76	+3.2
8	Byrum	J570W	0.75	5732	5280	6161	−7.89	+7.5
9	Ion Drive	K550W	0.79	8027	8642	7730	+7.66	−3.7
10	Caliber Isp 05 ARO-414 (Disc.)	I285	0.81	4930	4831	4772	−2.01	−3.2
11	Blister	K1075GG	0.83	9026	8347	8268	−7.52	−8.4
12	Caliber Isp 05 ARO-414 (Col.)	I285	0.84	5085	4842	4777	−4.78	−6.1
13	Rabia – Short Fin Can	L730	0.86	10584	10376	9916	−1.97	−6.3
14	Raven	J570W	1.07	8815	9288	9489	+5.37	+7.6
15	Rabia	L1080BB	1.14	12745	12777	11913	+0.25	−6.5
16	Torrent	M1850GG	1.22	12807	13852	12455	+8.16	−2.8
17	Kline-Rogers L500	L500	1.98	24771	26485	24179	+6.92	−2.4
18	A-601 Kinsel	P4935	2.19	42771	41086	46499	−3.94	+8.7
19	Full Metal Jacket – BALLS 005	O10000	2.31	37981	38820	37256	+2.21	−1.9
20	Full Metal Jacket – Black Rock 6	O10000	2.46	30038	32646	29239	+8.68	−2.7
21	Proteus 6	P9381	2.87	85067	86799	91339	+2.04	+7.4
22	AeroPac 104K Two-Stage	N1048 / M685W	3.04	104659	113786	103602	+8.72[†]	−1.0
23	Don't Debate This	N5800	3.04	56573	62308	53150	+10.14	−6.1
24	Qu8k	Q18000	3.46	121478	116254	119187	−4.30	−1.9
25	MESOS 293K	O4374 / M787	4.33	293488	289789	273056	−1.26[†]	−6.96

[†]For the two-stage flights AeroPac 104K and MESOS 293K, the recorded RASAero II apogees are post-flight simulations in which the ignition delay and launch angle were adjusted to match the measured flight; these RASAero II errors are therefore not blind forward predictions.

Table 10 Full exploratory high-Mach sounding-rocket set (reported in its entirety; none of these flights enters any headline statistic of Sec. VIII). Signed apogee error is (predicted − measured)/measured. Rows marked † are simulation failures, not physical predictions.

Flight	Source	Peak M	Error (%)	Within ±10%
Within ±10% (well-characterized vehicles)				
Black Brant V VB (AAF-VB-32)	[47]	7.224	−6.97	yes
Nike–Deacon no. 1 (Wallops 1955)	[48]	4.956	−1.06	yes
Nike–Deacon no. 2 (Wallops 1955)	[48]	5.079	−0.89	yes
Over-predicted (Nike–Apache 1965 set; Nike–Cajun)				
Nike–Apache 14.108 GI	[49]	6.498	31.47	no
Nike–Apache 14.75 GR (Wallops 1965)	X-721-67-103	6.747	24.49	no
Nike–Apache 14.78 UA (WSMR 1965)	X-721-67-103	5.955	23.87	no
Nike–Apache 14.133 NA (Wallops 1965)	X-721-67-103	6.335	33.82	no
Nike–Apache 14.213 UI (Wallops 1965)	X-721-67-103	6.426	35.85	no
Nike–Apache 14.214 UI (Wallops 1965)	X-721-67-103	6.416	35.32	no
Nike–Apache 14.242 UE (Wallops 1965)	X-721-67-103	7.047	32.68	no
Nike–Apache 14.244 UI (Wallops 1965)	X-721-67-103	6.939	26.88	no
Nike–Apache 14.247 UI (Wallops 1965)	X-721-67-103	7.047	29.63	no
Nike–Cajun UM (Wallops 1956–57)	NACA RM L57D26	6.249	16.56	no
Under-predicted (Arcas family; HEROS 3)				
Arcas flight 2 (blunt)	DTIC AD-235341	2.294	−29.18	no
Arcas flight 3 (blunt)	DTIC AD-235341	2.294	−38.76	no
Arcas flight 4 (secant ogive)	DTIC AD-235341	2.535	−69.12	no
Arcas flight 5 (secant ogive)	DTIC AD-235341	2.535	−67.93	no
HEROS 3 (Esrange 2016)	Kobald 2018	1.886	−63.38	no
Simulation failures (no physical prediction)				
Nike–Apache 14.210 GI†	X-721-67-103	—	zero apogee	no
Nike–Cajun Hurricane†	NACA RM L57D26	—	integration error	no

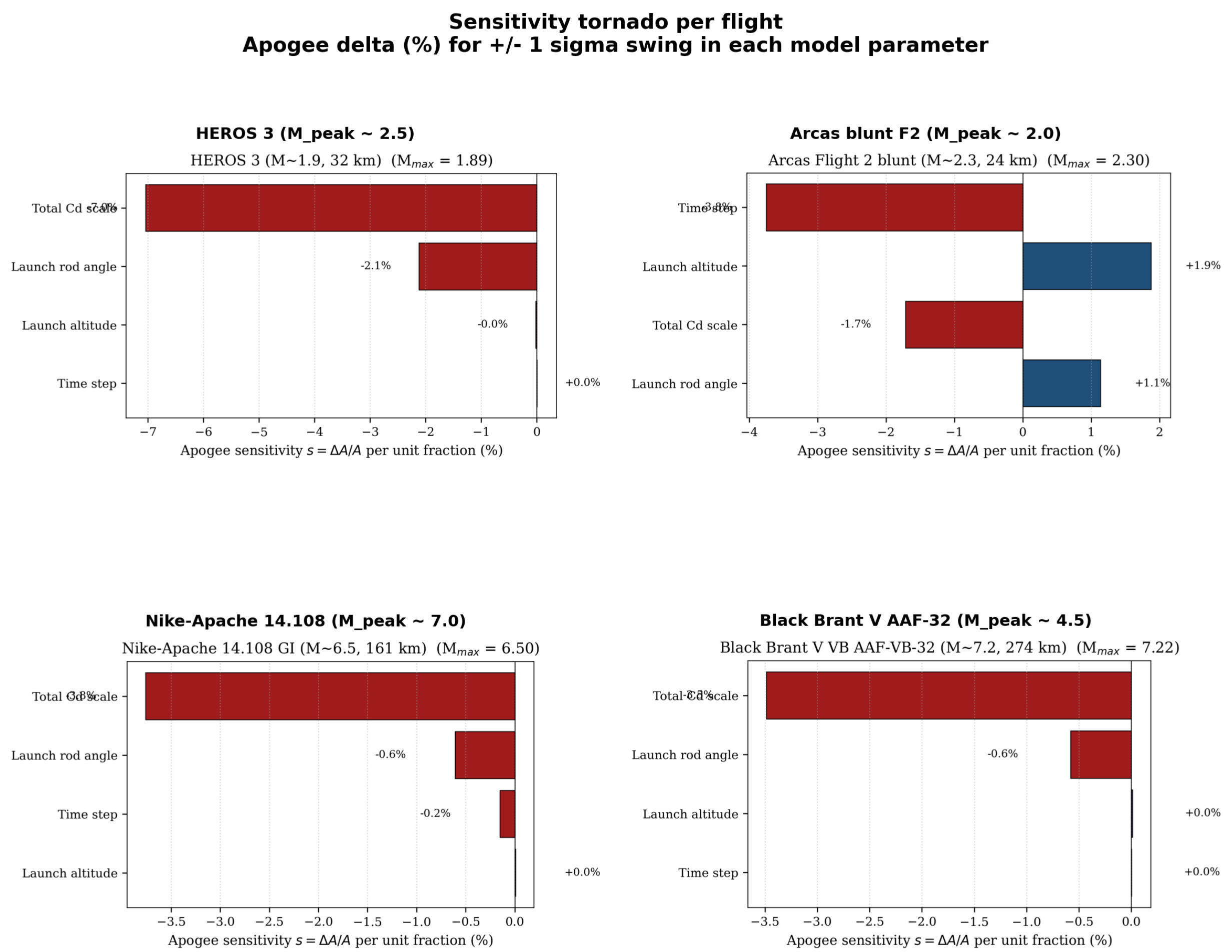

Fig. 26 Sensitivity tornado composite for the four high-Mach exploratory vehicles (HEROS 3, blunt Arcas, Nike-Apache, Black Brant VB) under ±10% perturbations of drag-coefficient scale, integration time step, launch altitude, and launch-rod angle. These worst-case trajectories bound the numerical-convergence behavior of the gentler headline flights; drag-coefficient scale dominates the response.